

BFS Traversal

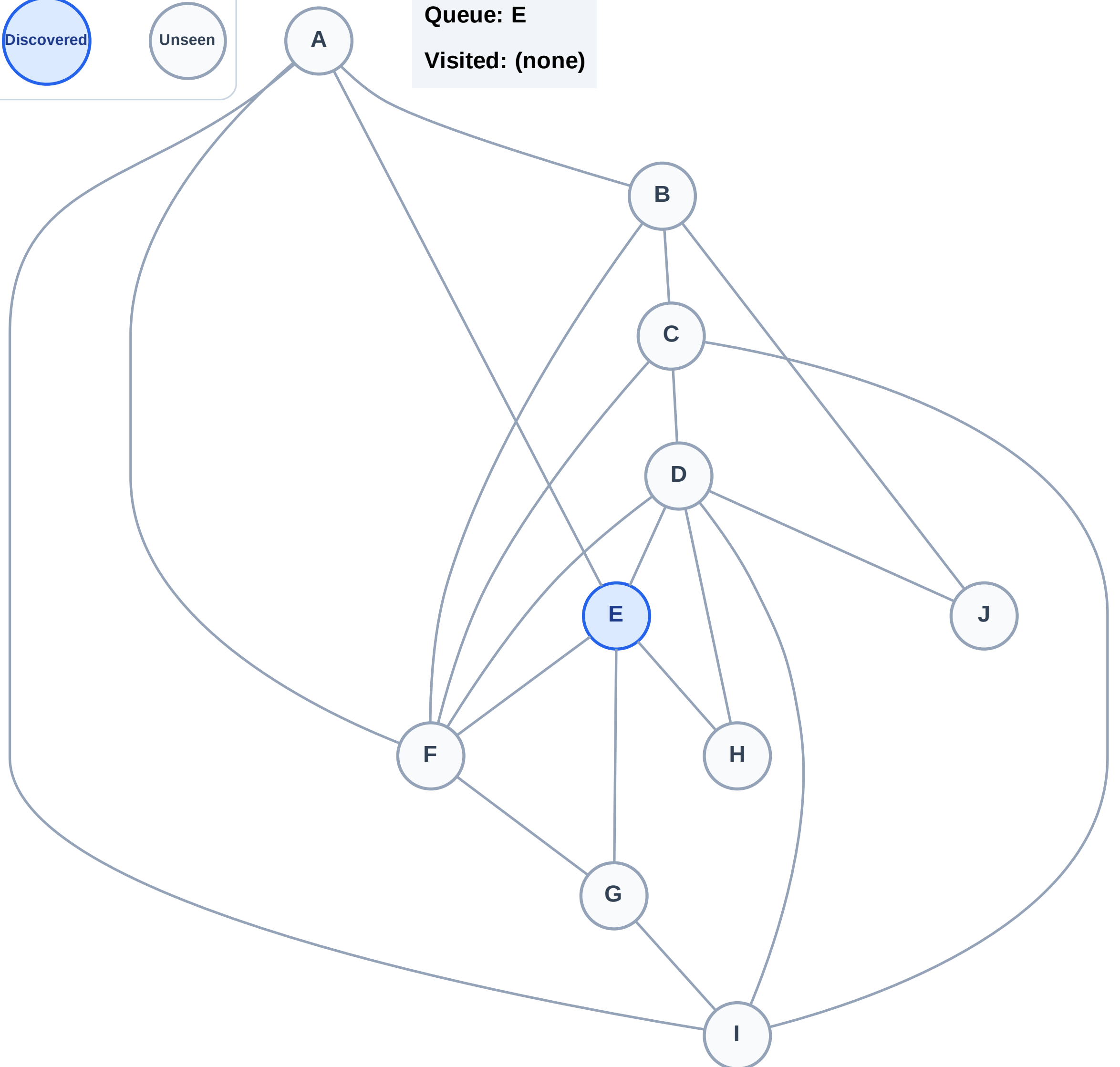
Step 1: Start at E

Legend



Queue: E

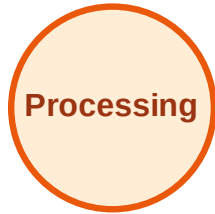
Visited: (none)



BFS Traversal

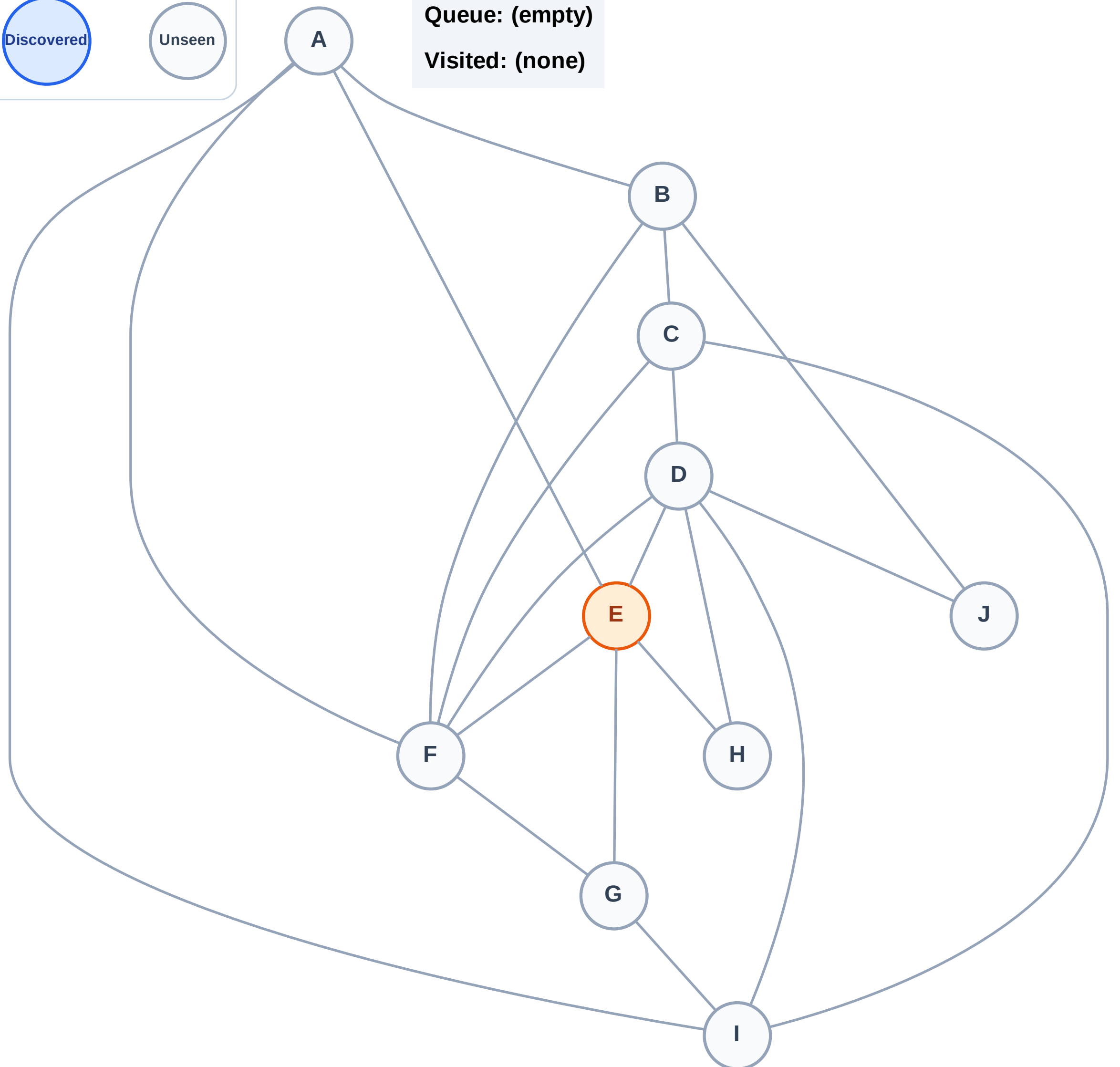
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

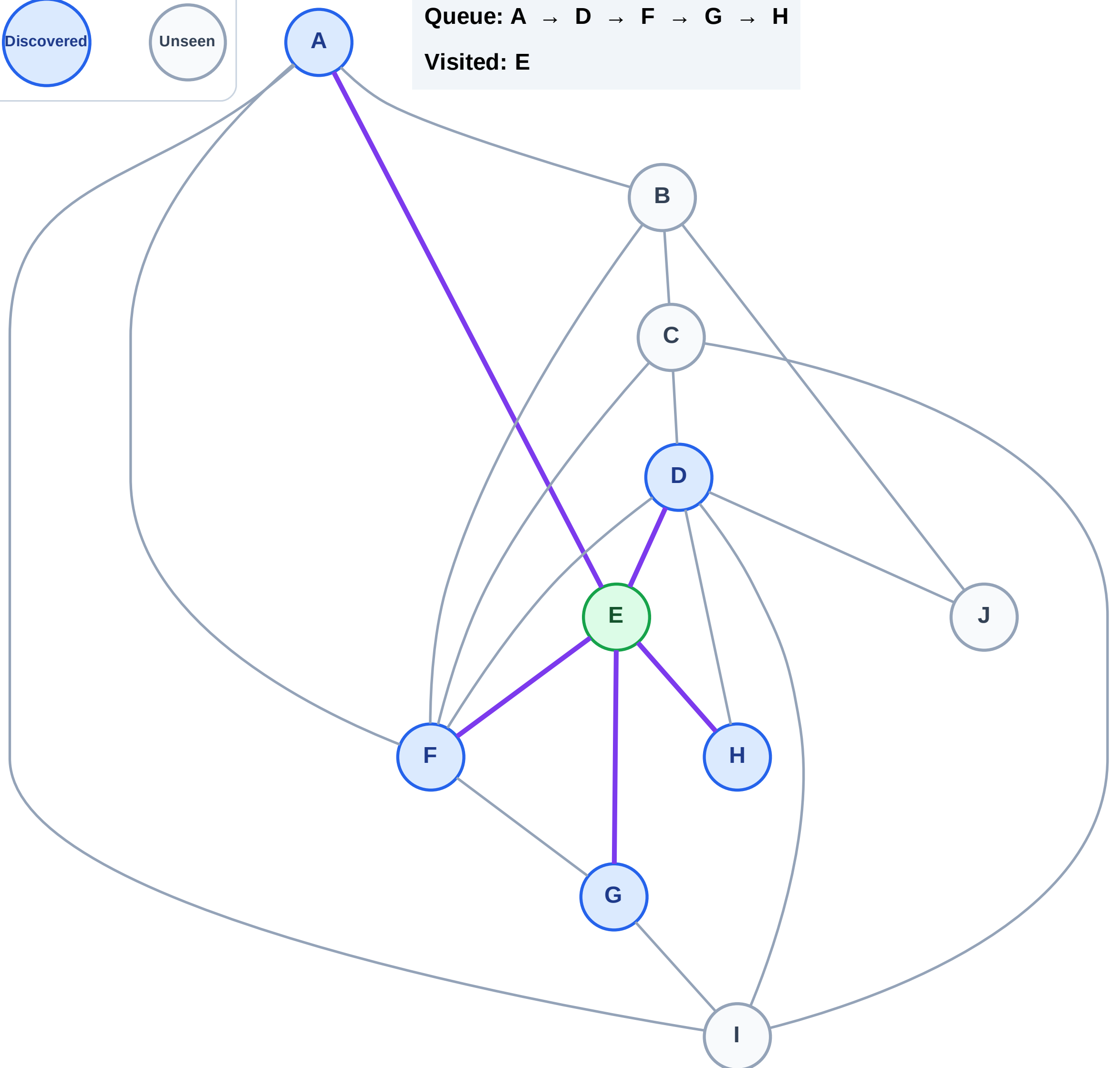
Step 3: Discover A, D, F, G, H from E

Legend



Queue: A → D → F → G → H

Visited: E



BFS Traversal

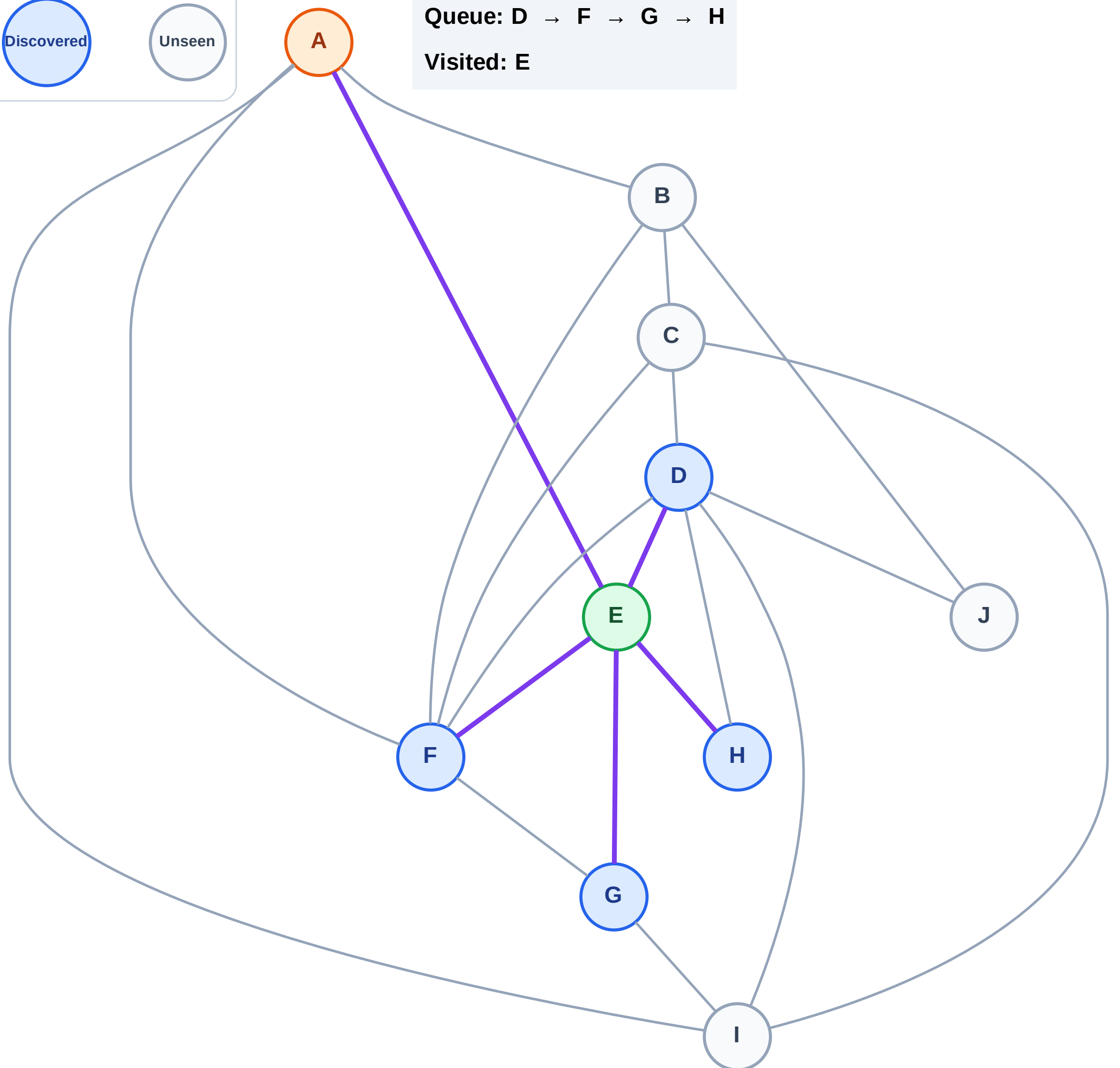
Step 4: Process node A

Legend



Queue: D → F → G → H

Visited: E



BFS Traversal

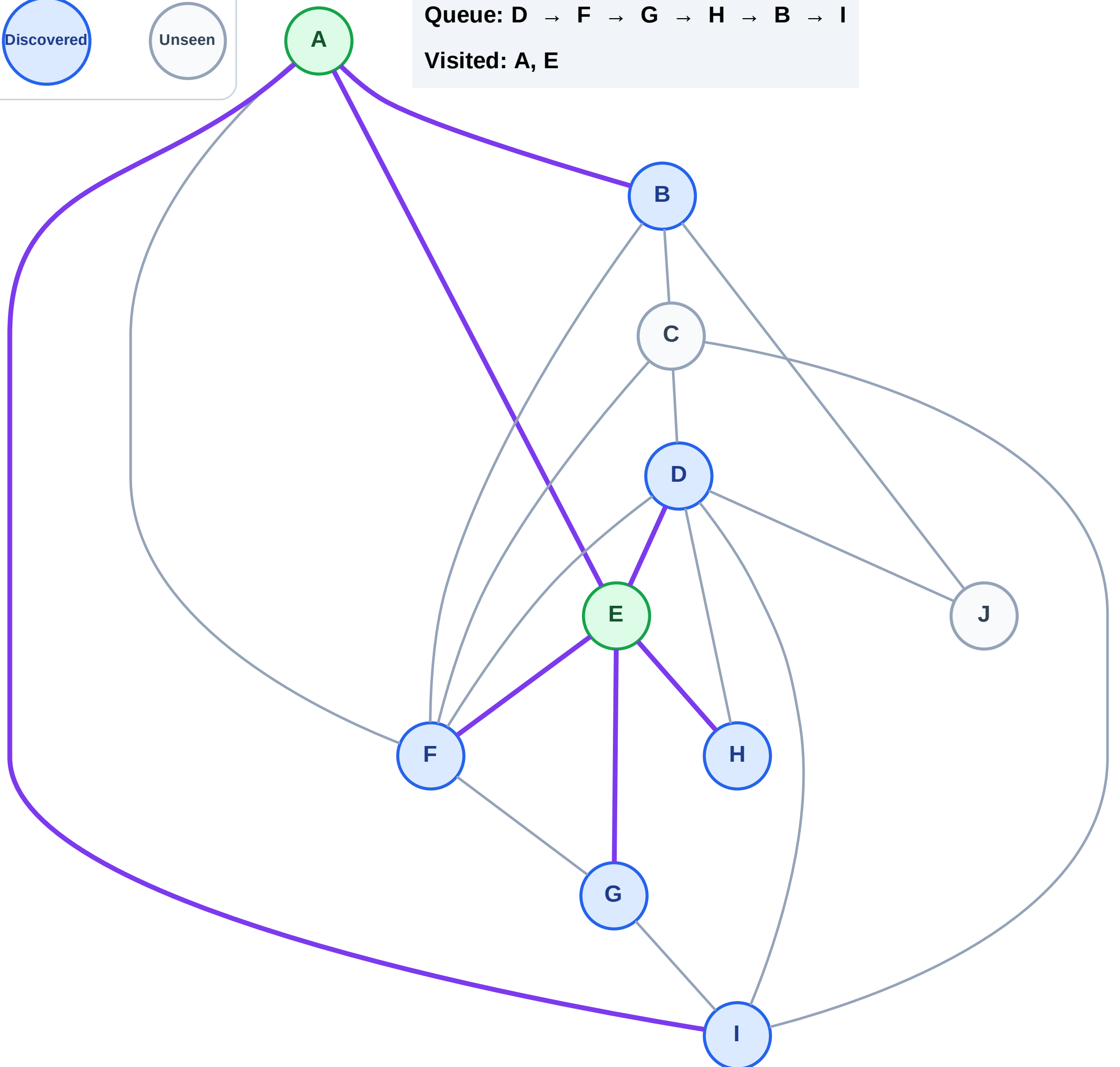
Step 5: Discover B, I from A

Legend



Queue: D → F → G → H → B → I

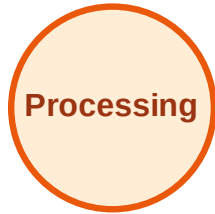
Visited: A, E



BFS Traversal

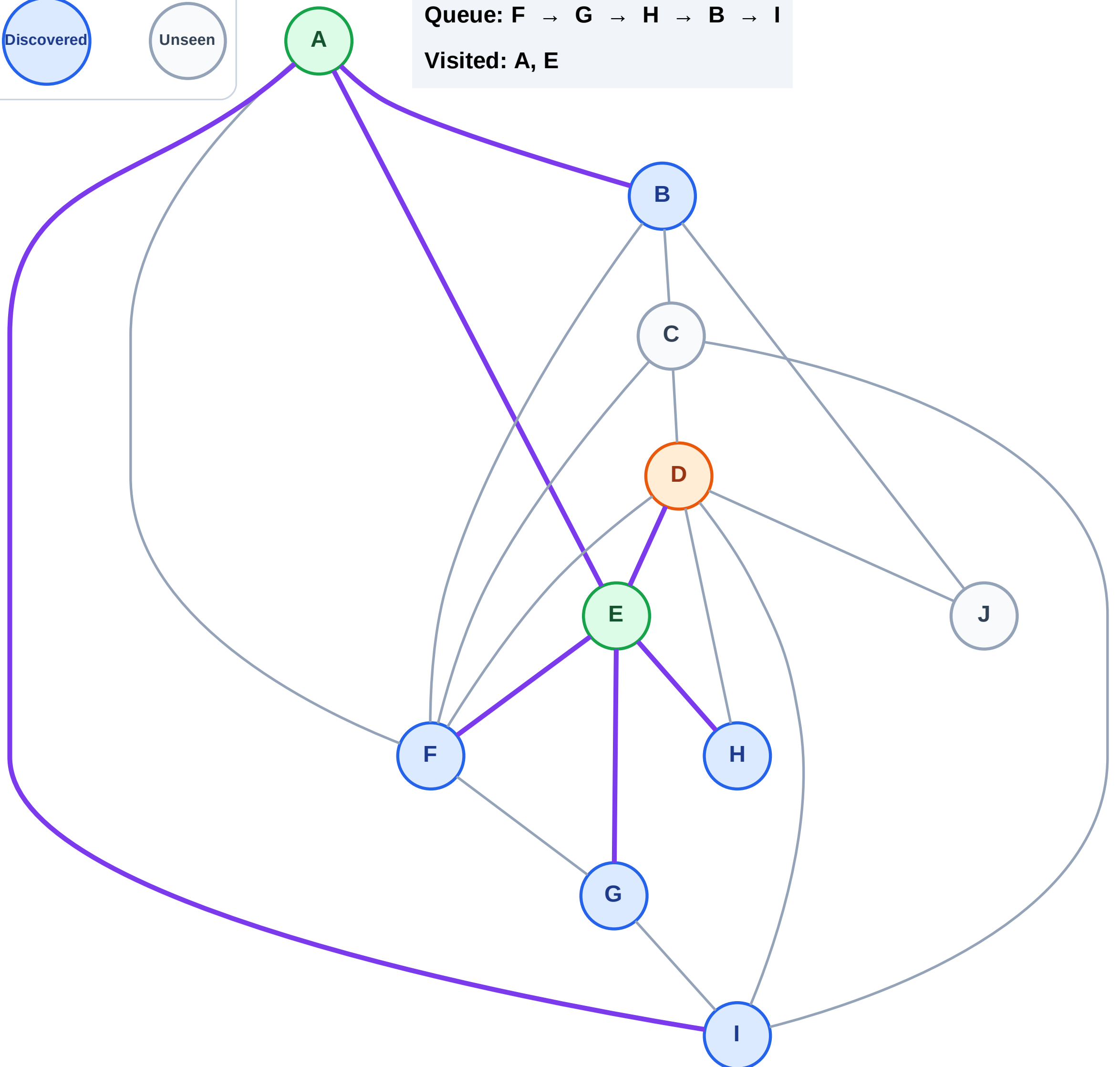
Step 6: Process node D

Legend



Queue: F → G → H → B → I

Visited: A, E



BFS Traversal

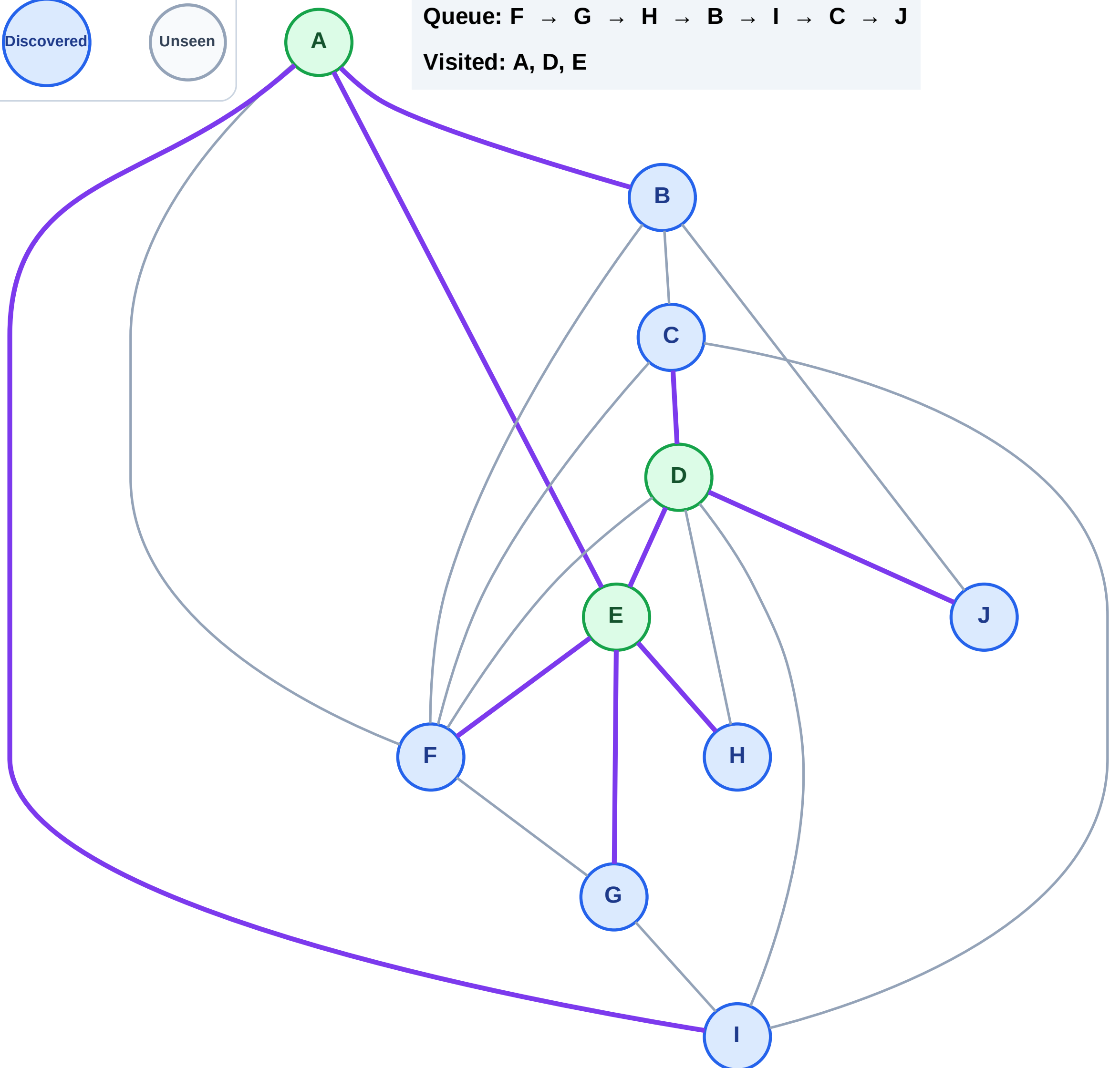
Step 7: Discover C, J from D

Legend



Queue: F → G → H → B → I → C → J

Visited: A, D, E



BFS Traversal

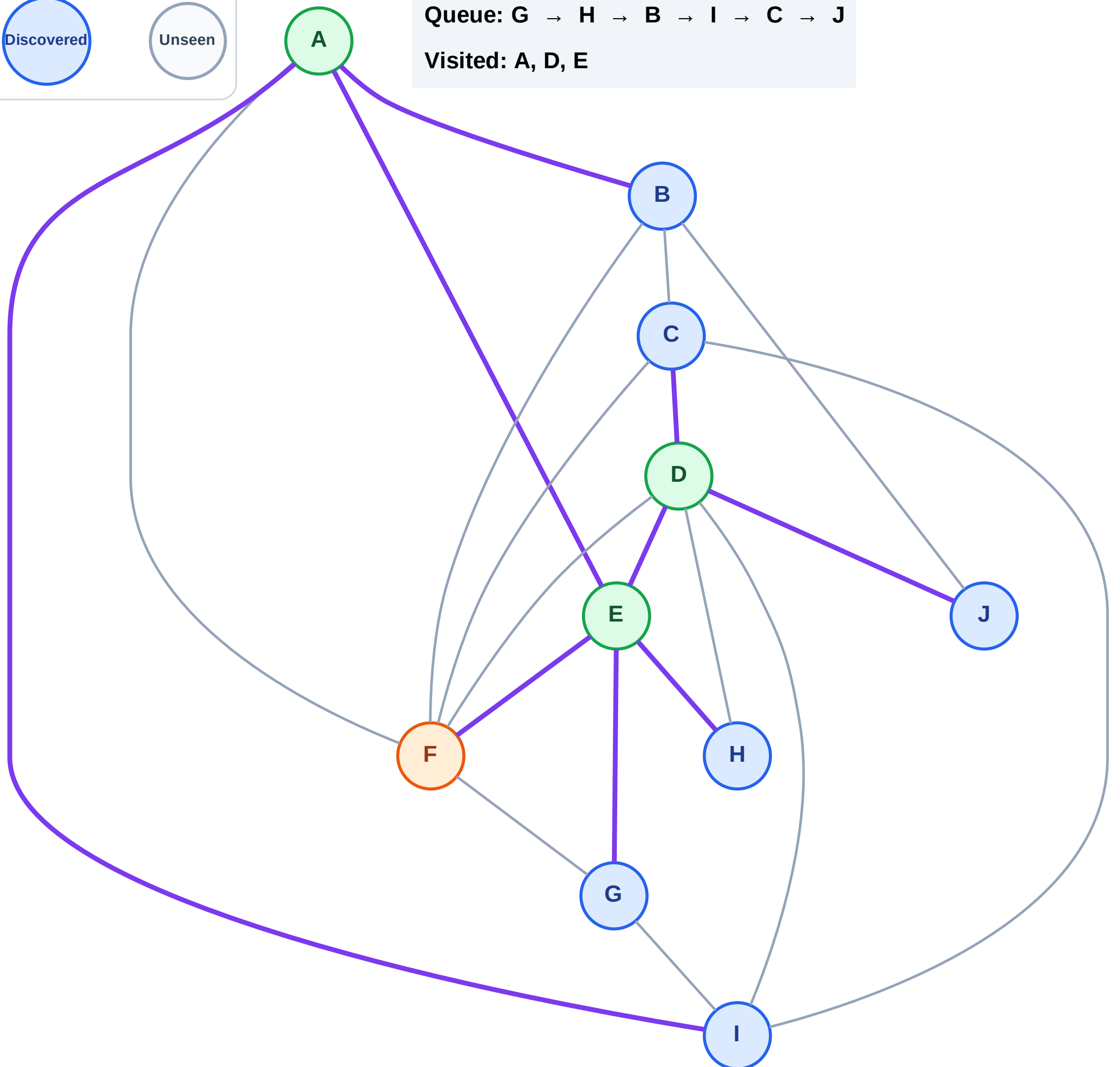
Step 8: Process node F

Legend



Queue: G → H → B → I → C → J

Visited: A, D, E



BFS Traversal

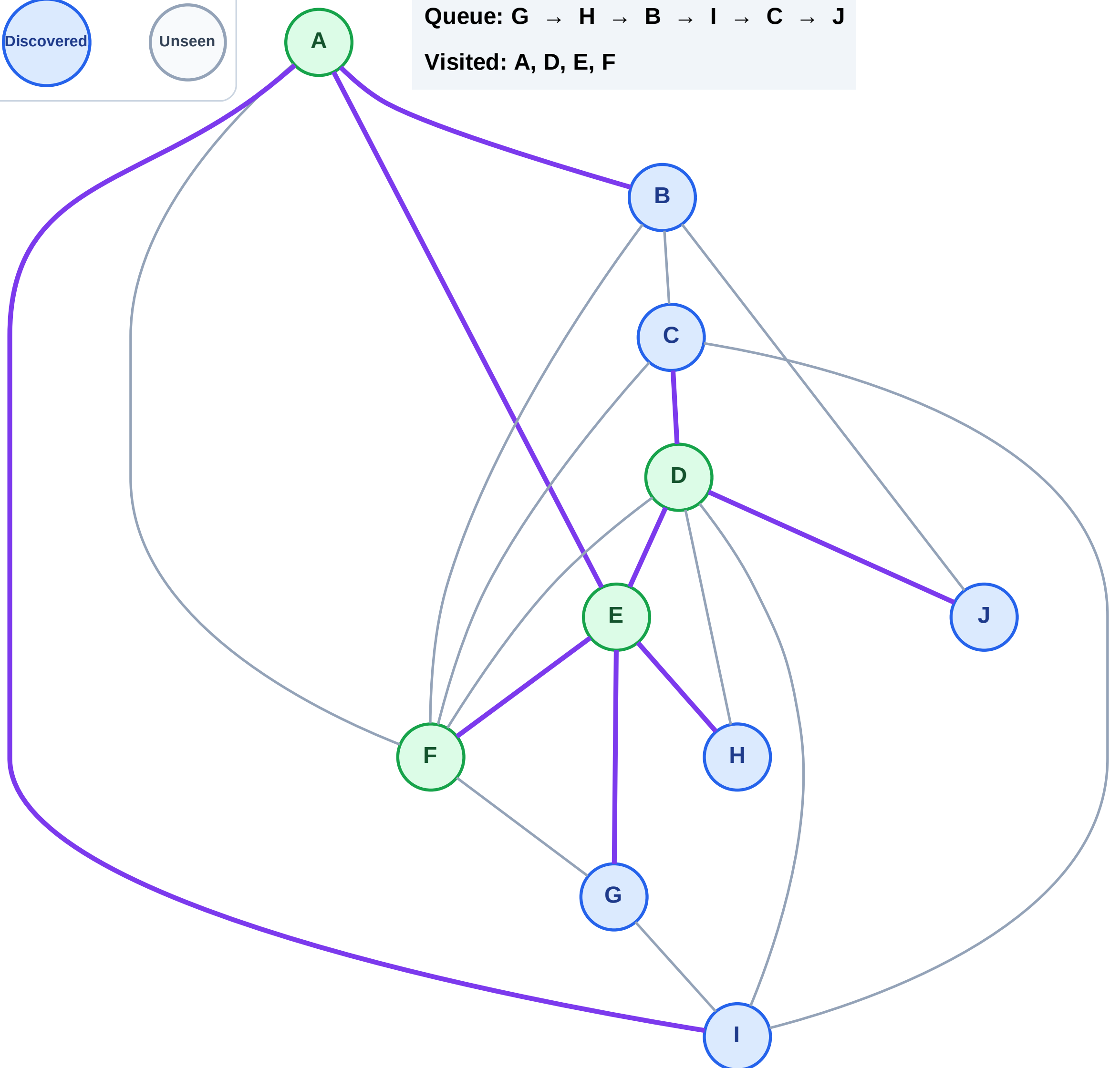
Step 9: No new neighbors from F

Legend



Queue: G → H → B → I → C → J

Visited: A, D, E, F



BFS Traversal

Step 10: Process node G

Legend

Complete

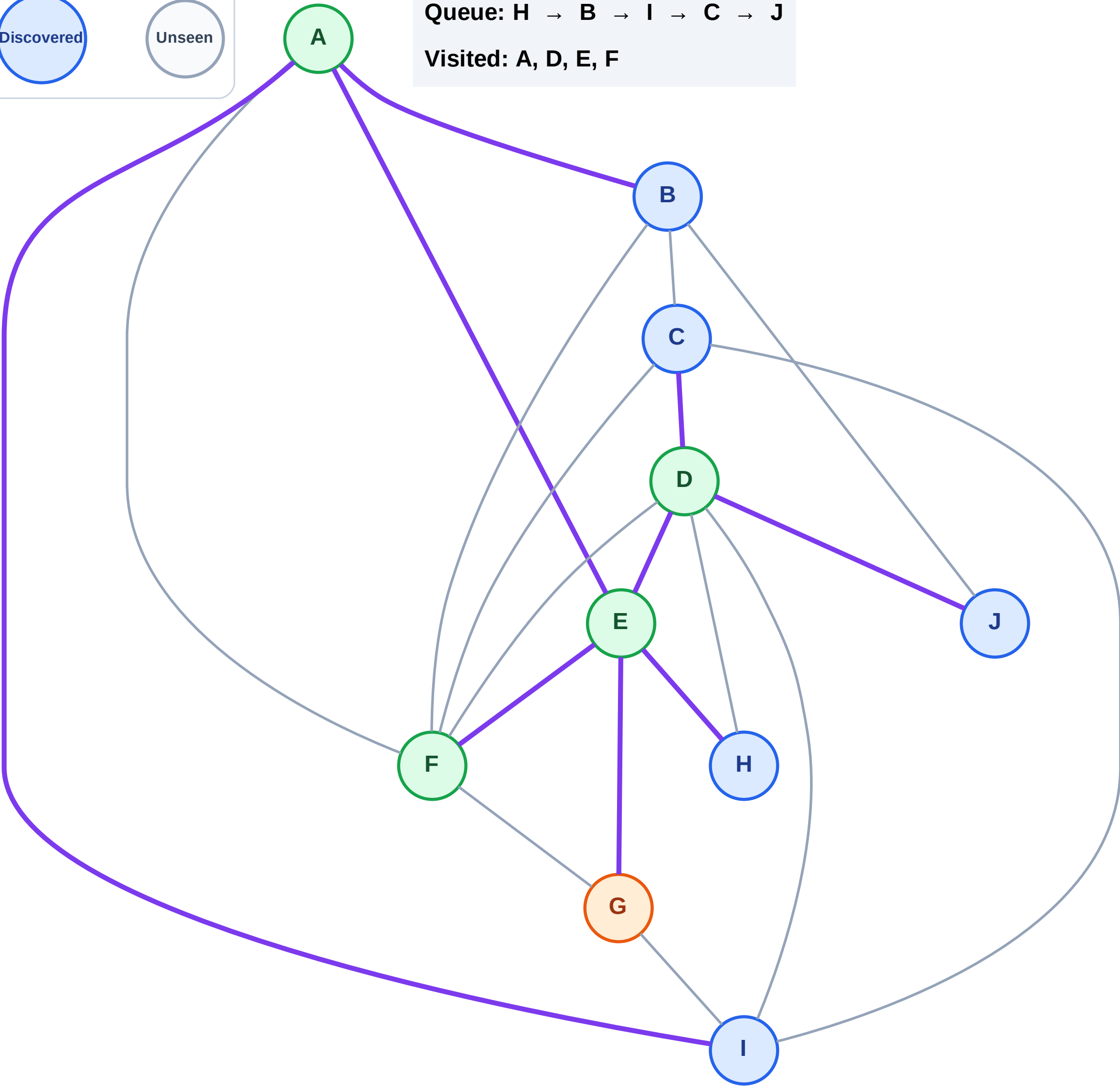
Processing

Discovered

Unseen

Queue: H → B → I → C → J

Visited: A, D, E, F



BFS Traversal

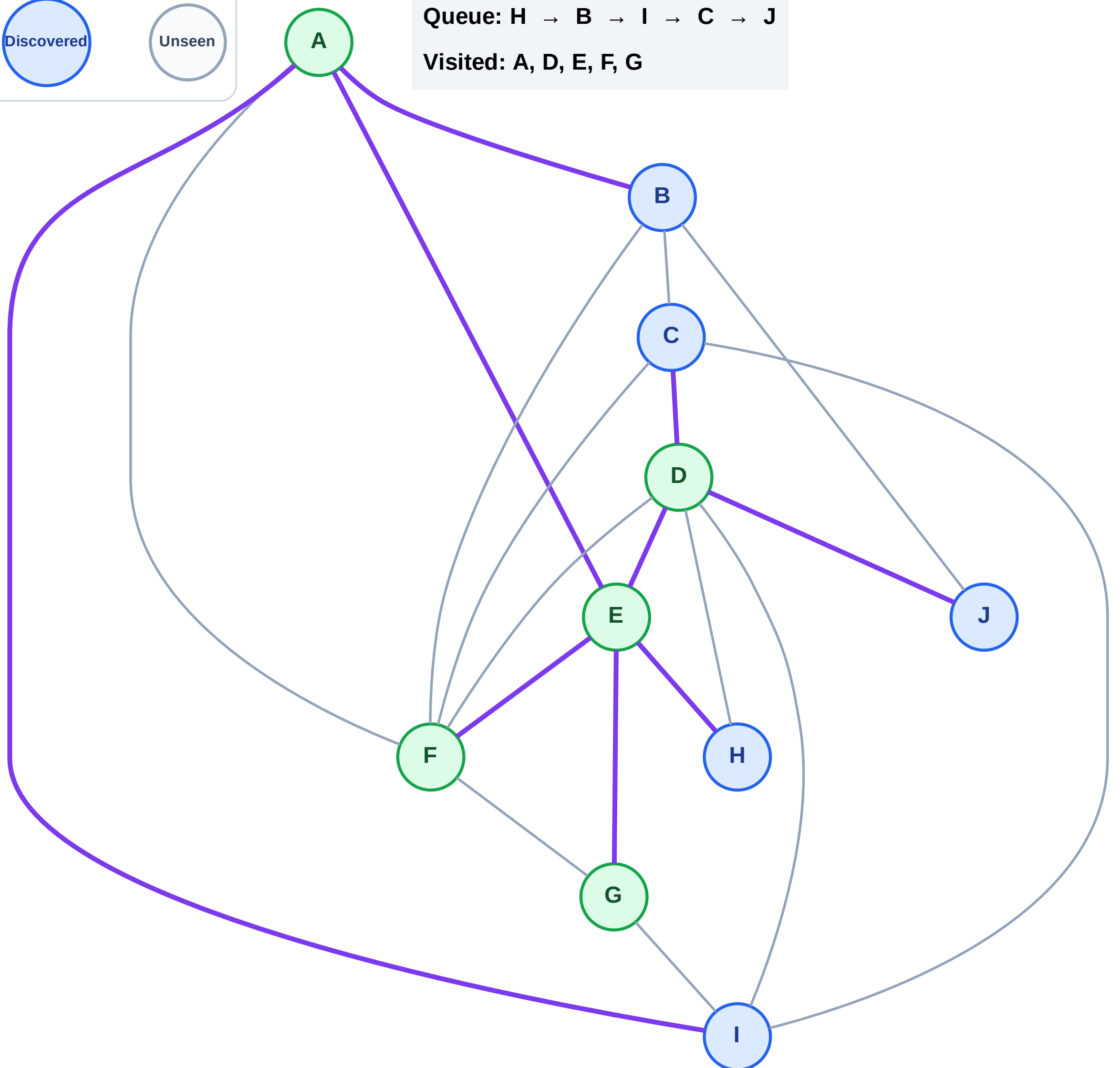
Step 11: No new neighbors from G

Legend



Queue: H → B → I → C → J

Visited: A, D, E, F, G



BFS Traversal

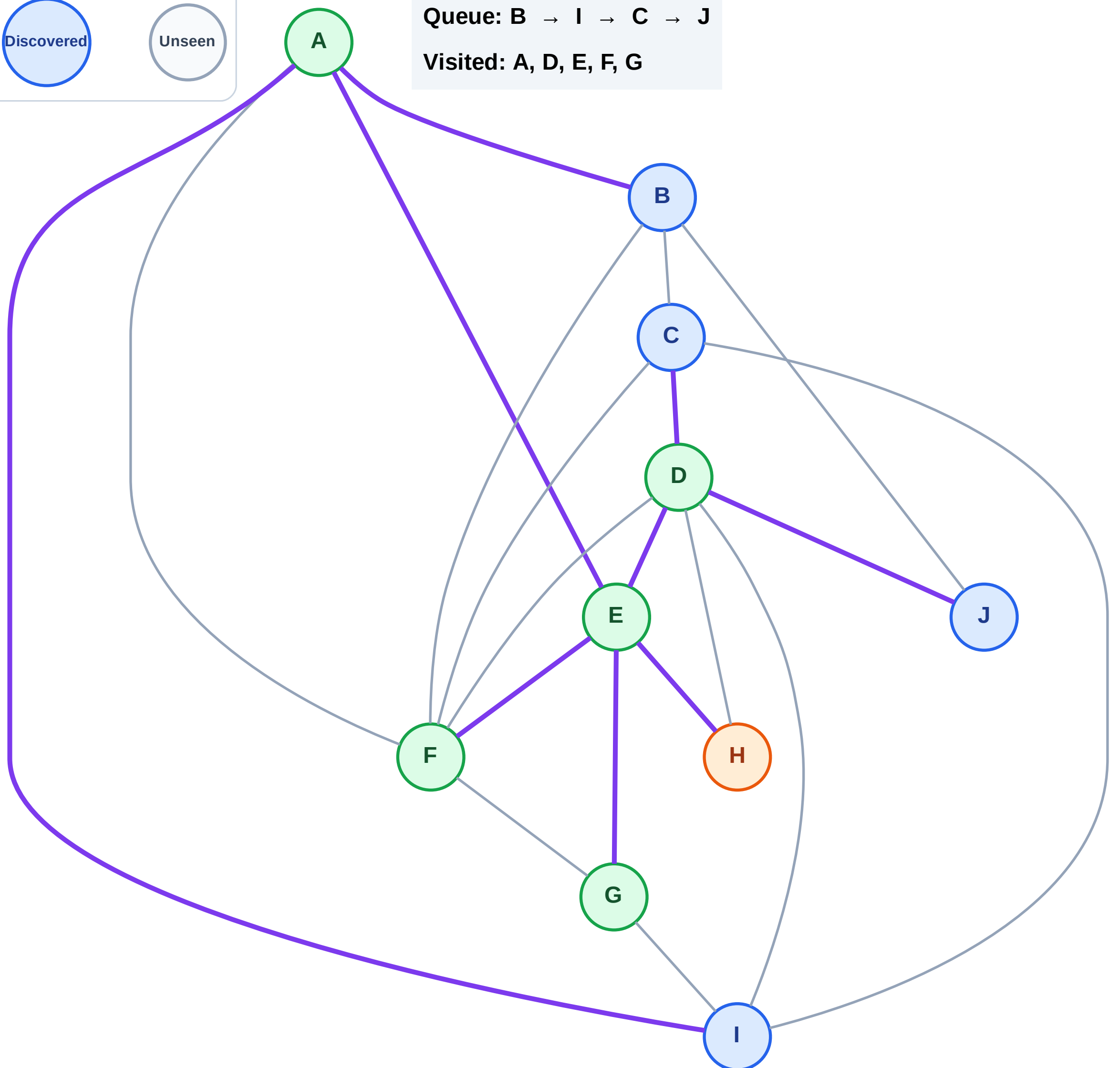
Step 12: Process node H

Legend



Queue: B → I → C → J

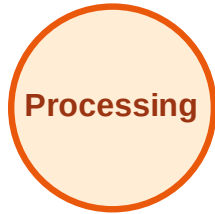
Visited: A, D, E, F, G



BFS Traversal

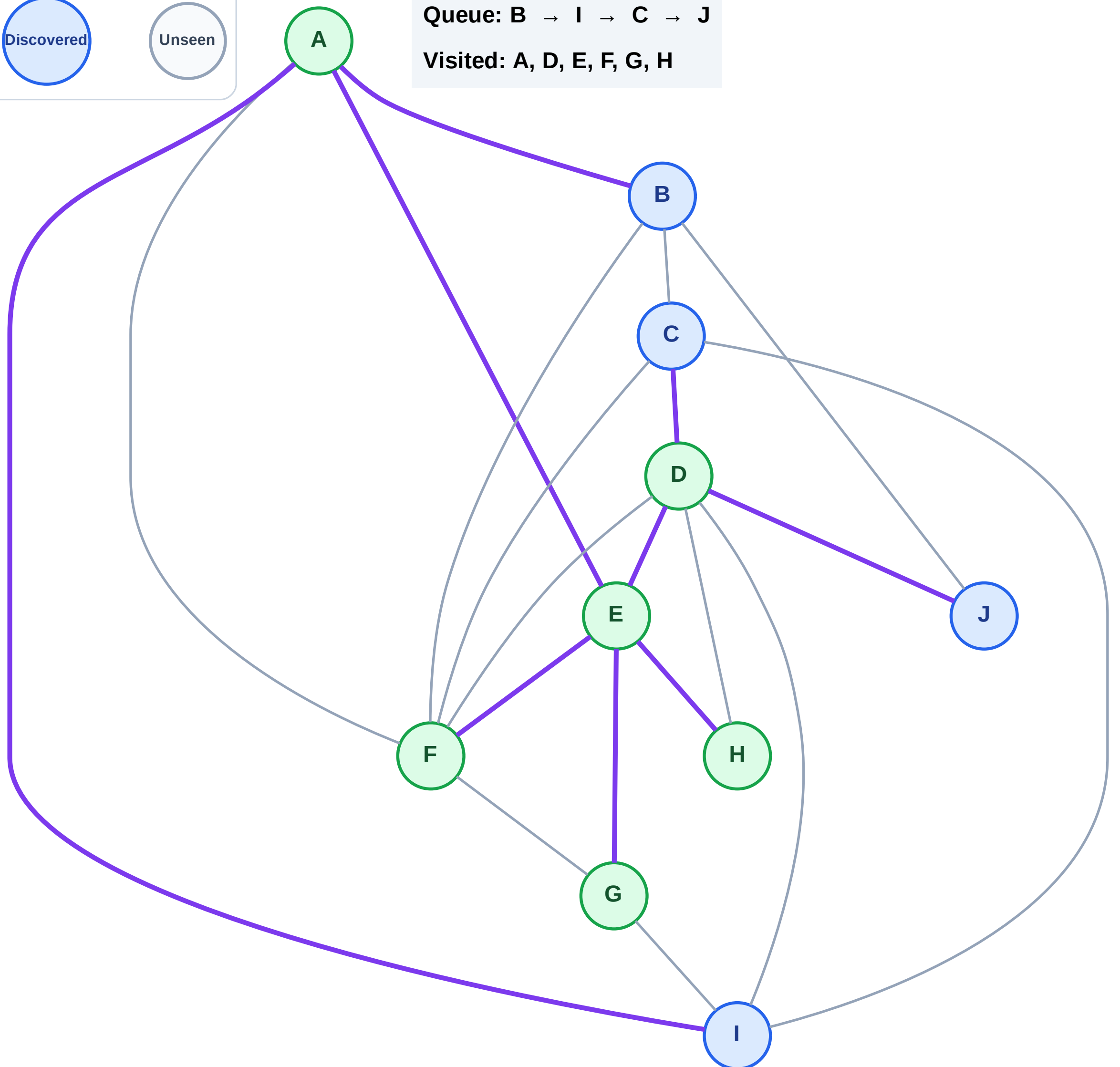
Step 13: No new neighbors from H

Legend



Queue: B → I → C → J

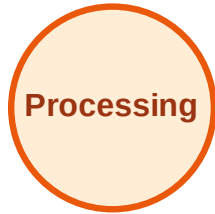
Visited: A, D, E, F, G, H



BFS Traversal

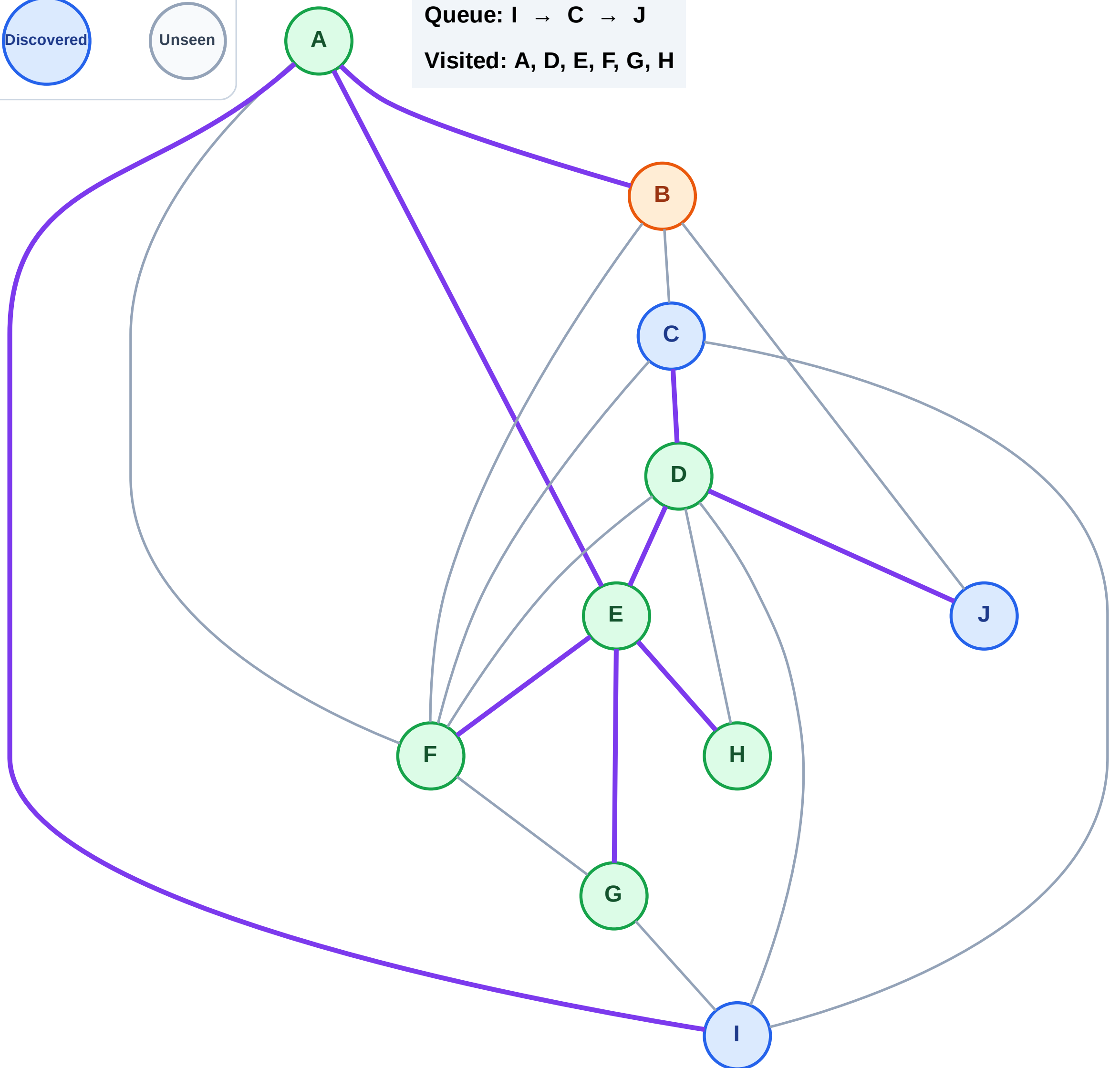
Step 14: Process node B

Legend



Queue: I → C → J

Visited: A, D, E, F, G, H



BFS Traversal

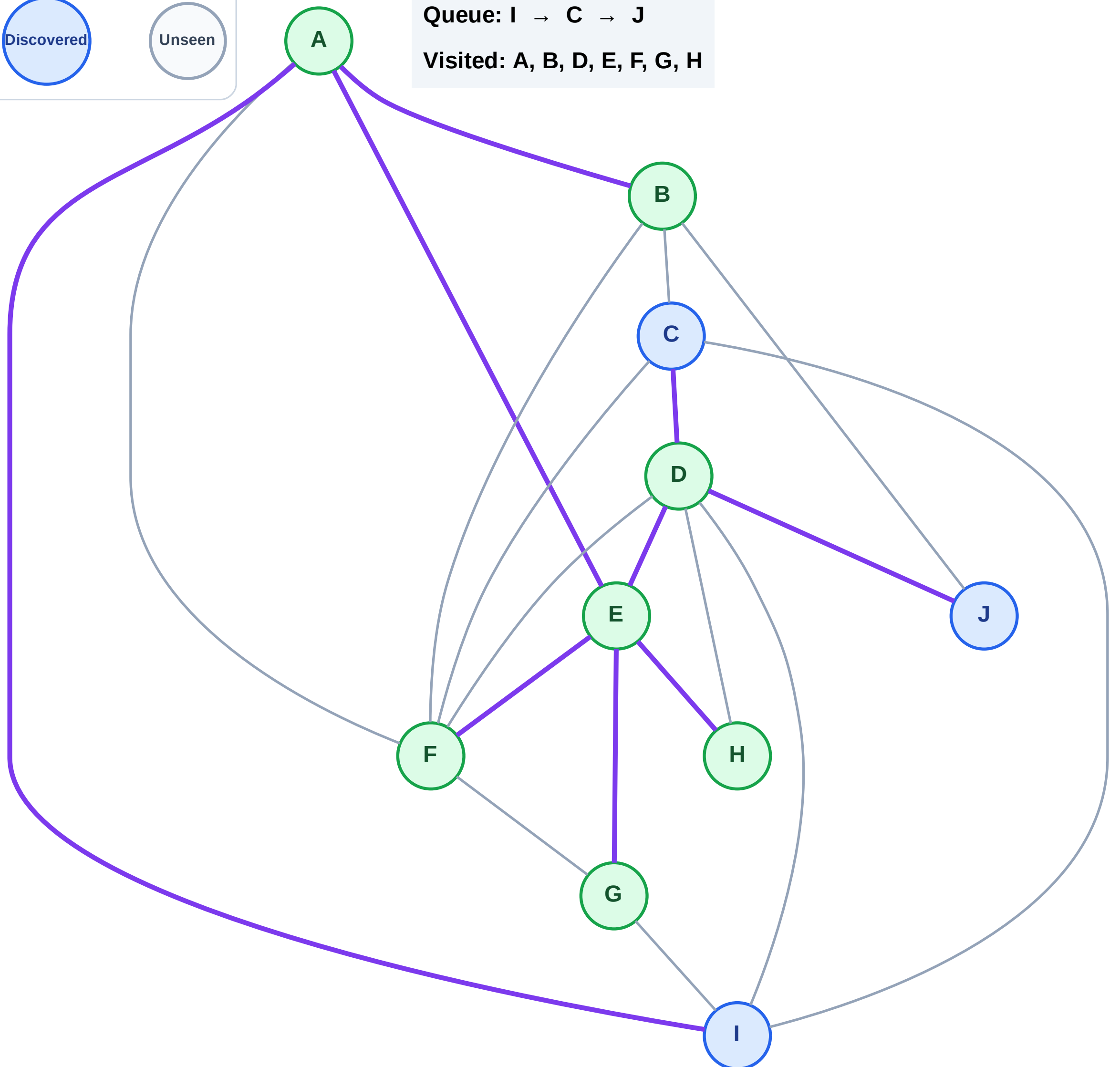
Step 15: No new neighbors from B

Legend



Queue: I → C → J

Visited: A, B, D, E, F, G, H



BFS Traversal

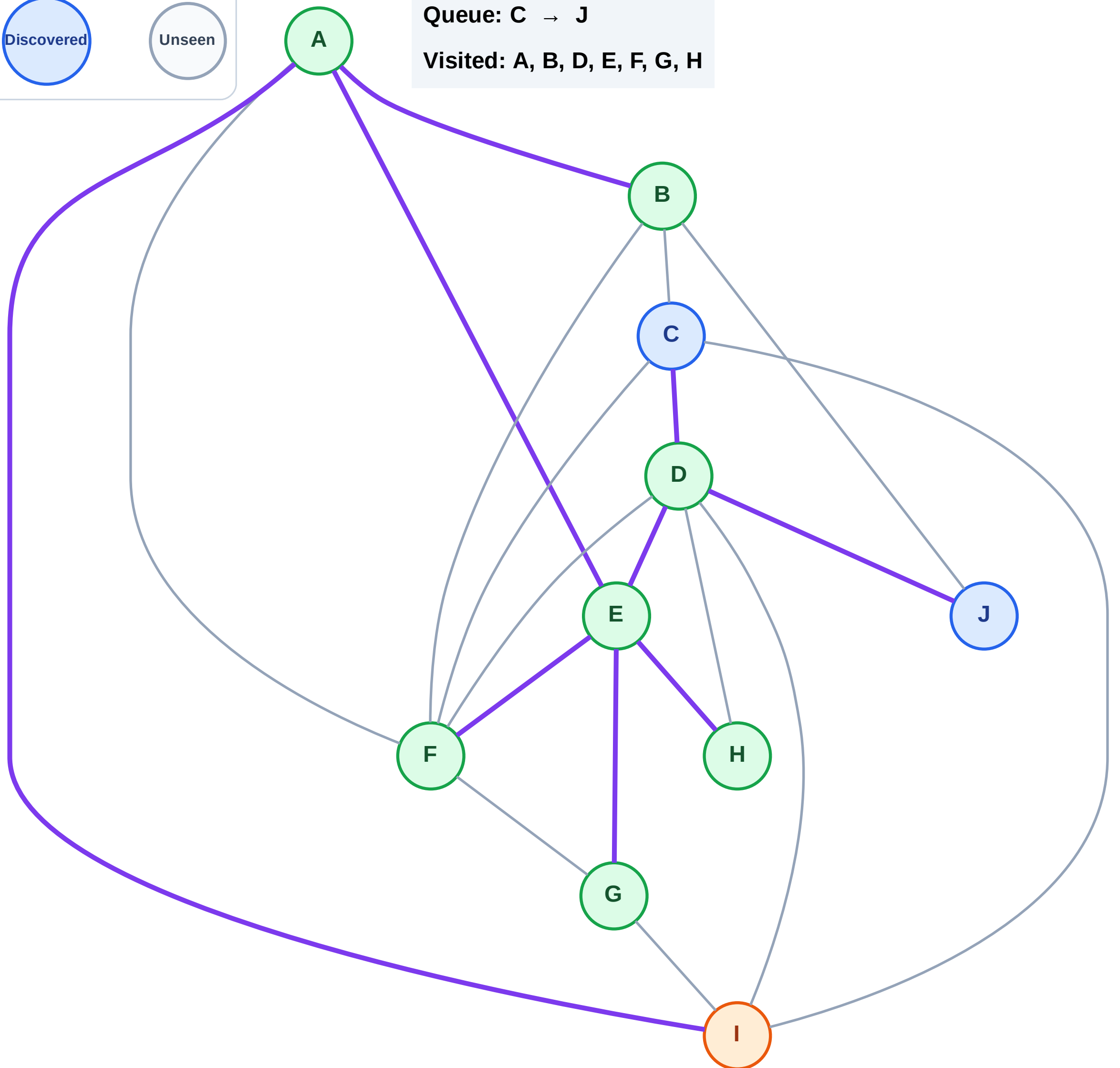
Step 16: Process node I

Legend



Queue: C → J

Visited: A, B, D, E, F, G, H



BFS Traversal

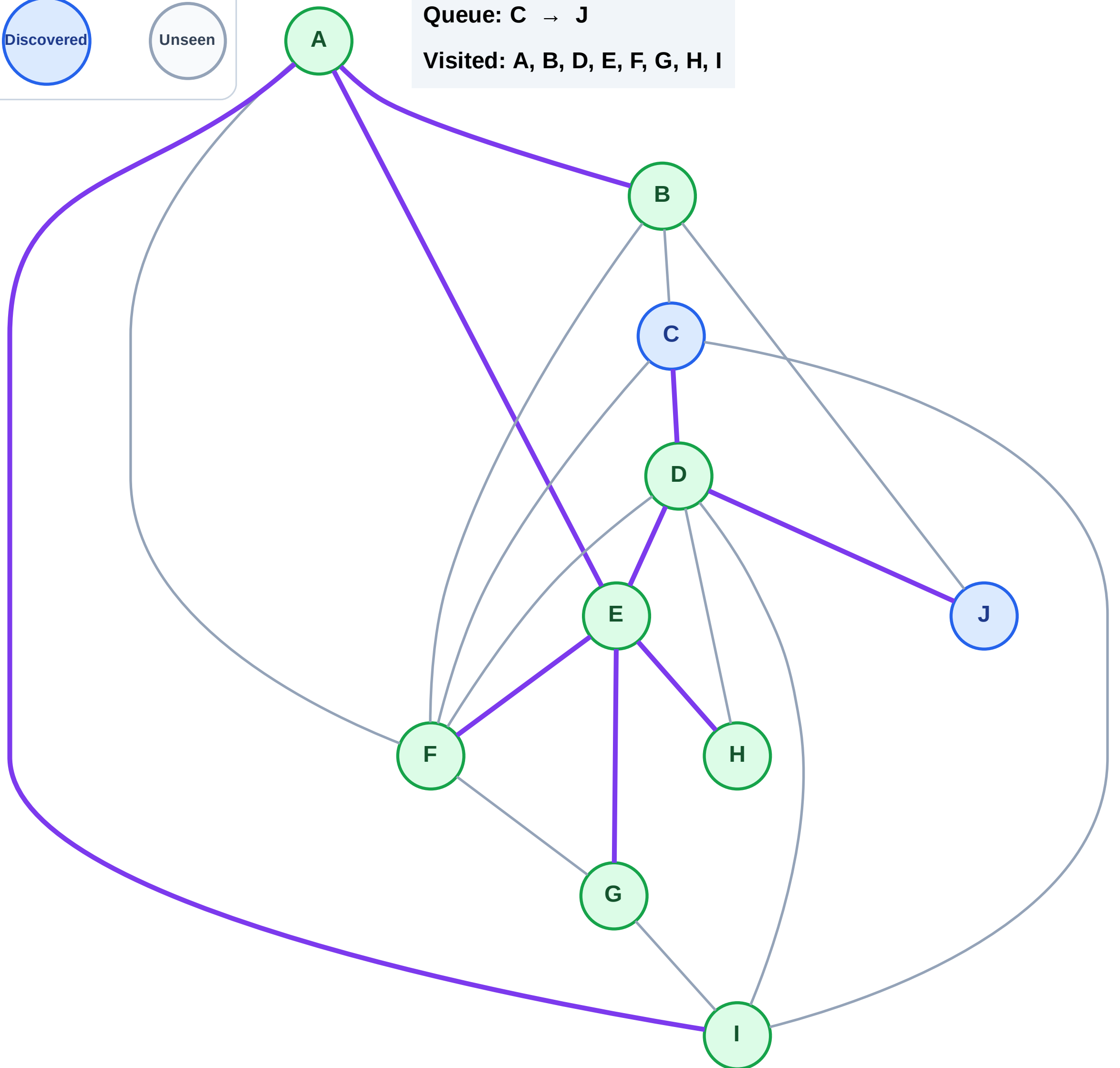
Step 17: No new neighbors from I

Legend



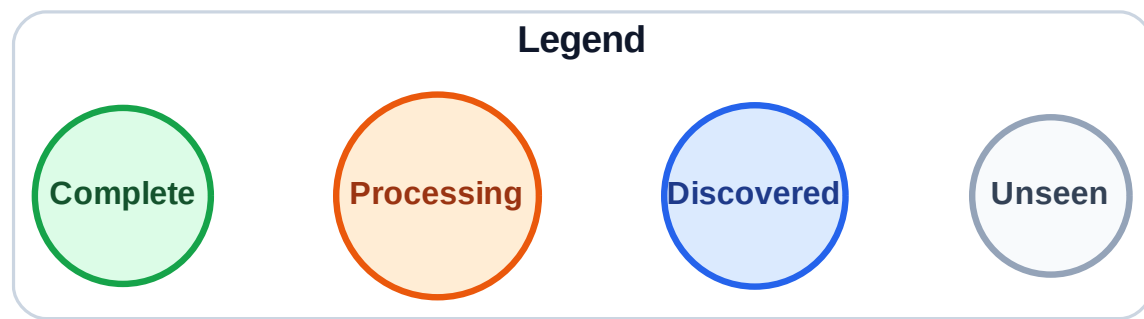
Queue: C → J

Visited: A, B, D, E, F, G, H, I

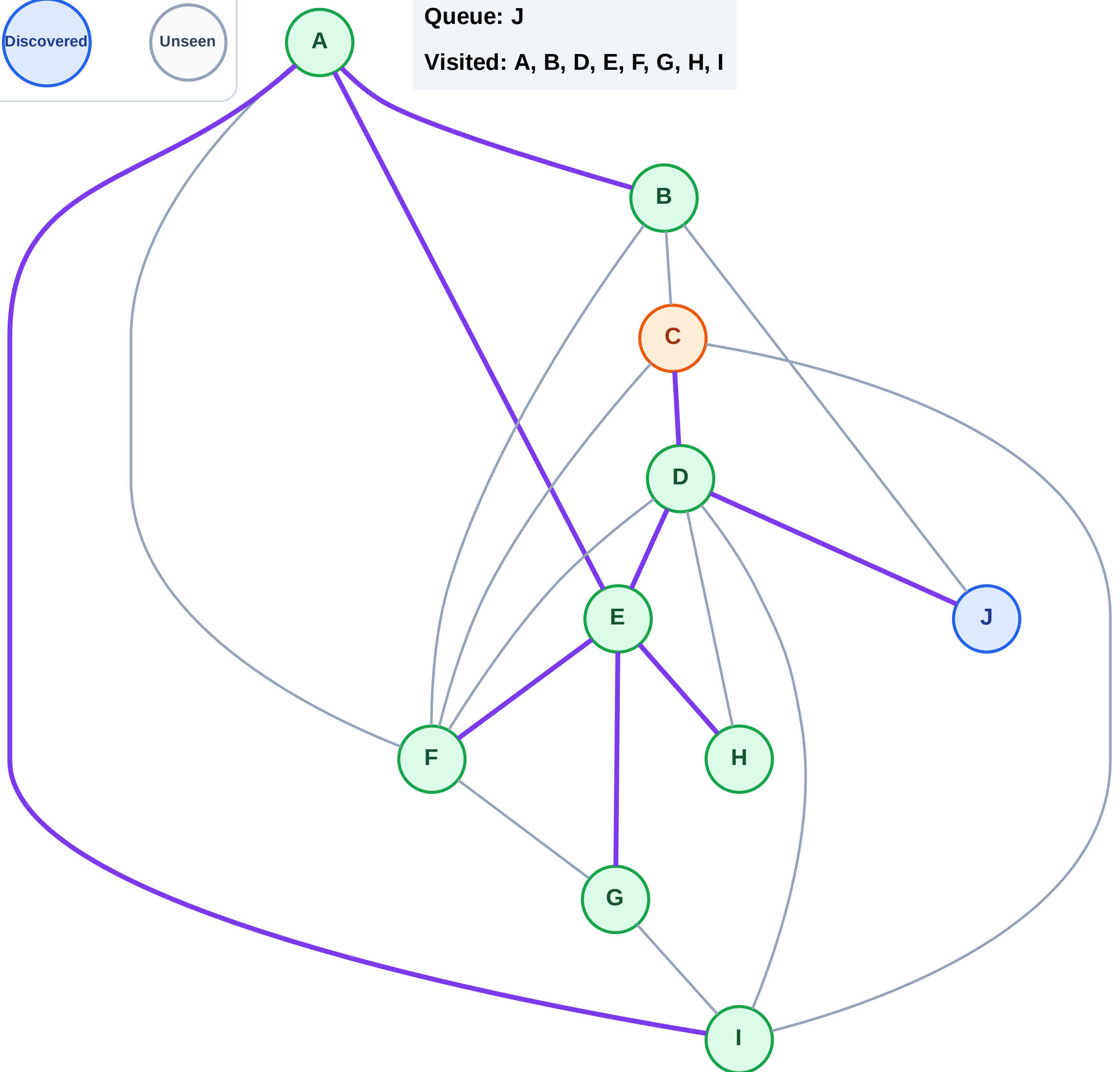


BFS Traversal

Step 18: Process node C



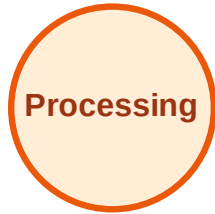
Queue: J
Visited: A, B, D, E, F, G, H, I



BFS Traversal

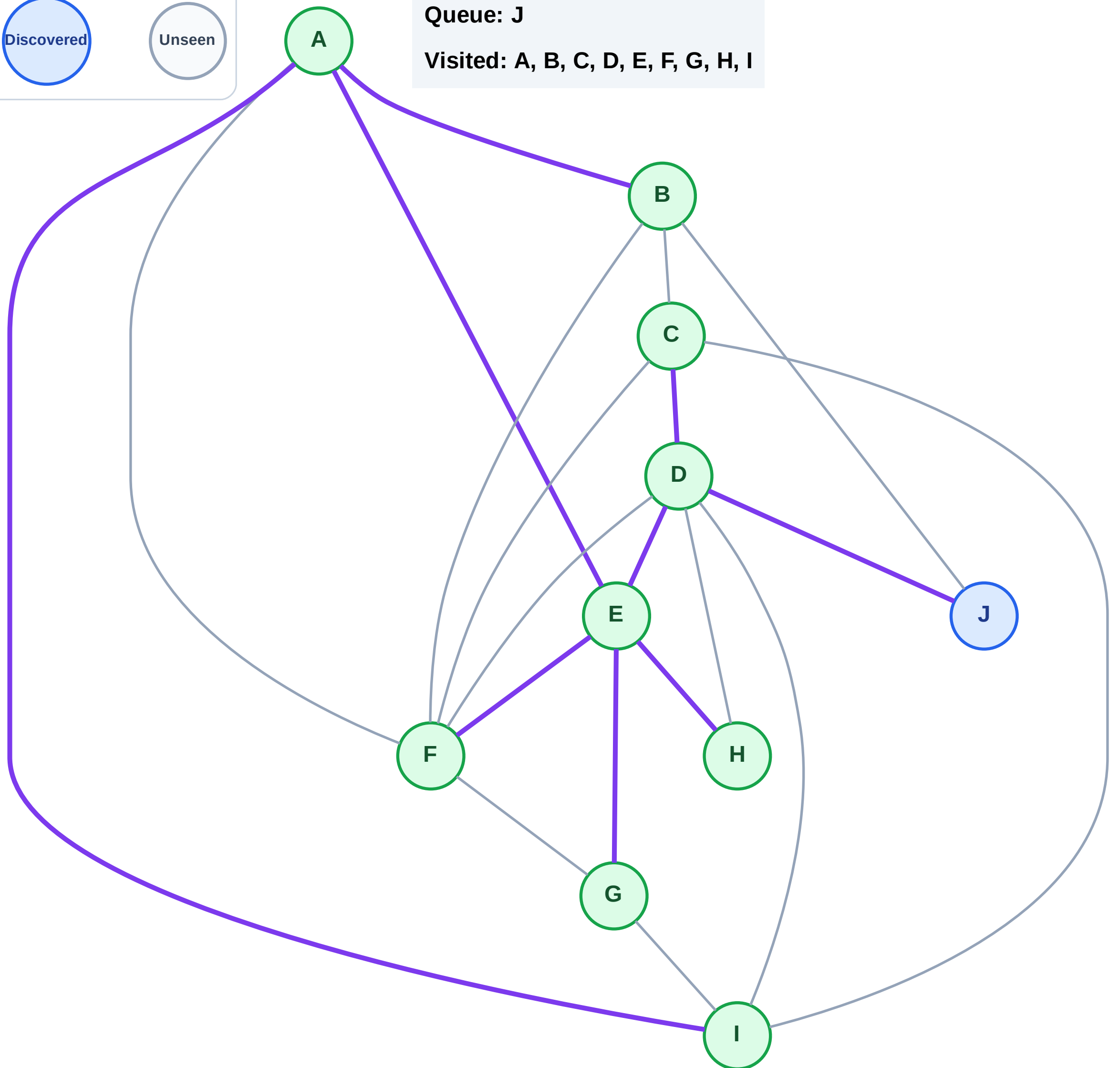
Step 19: No new neighbors from C

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

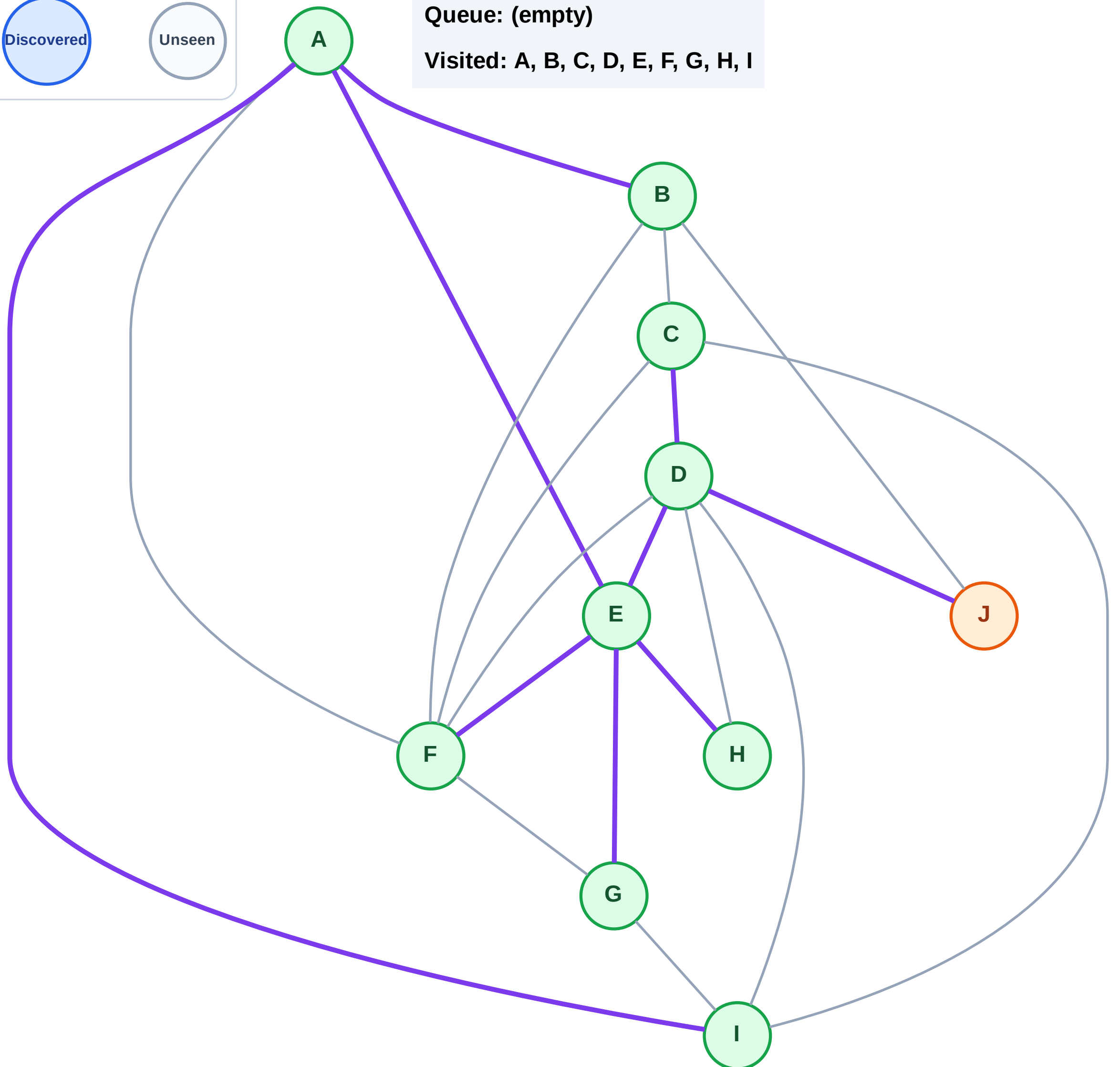
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

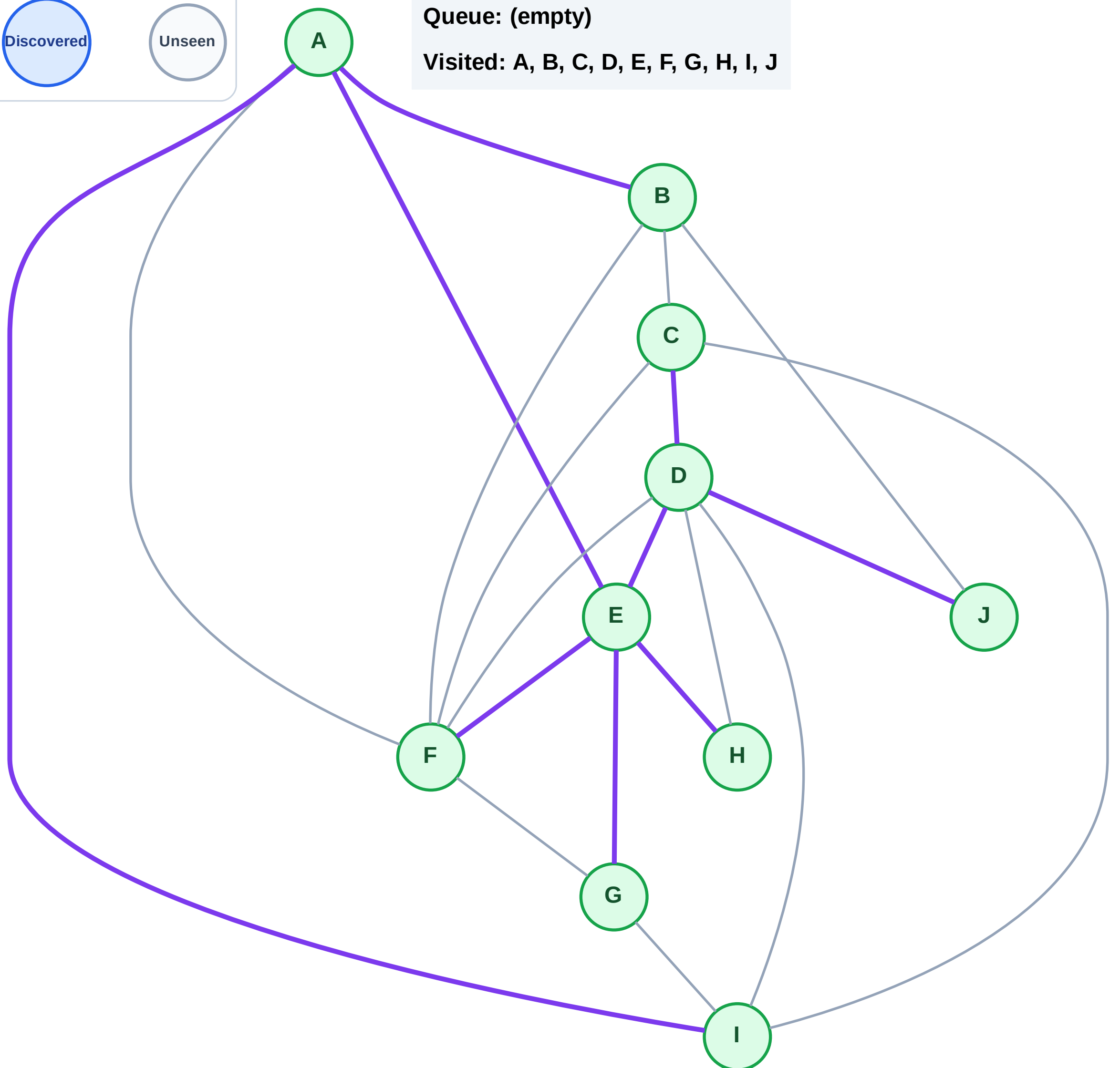
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

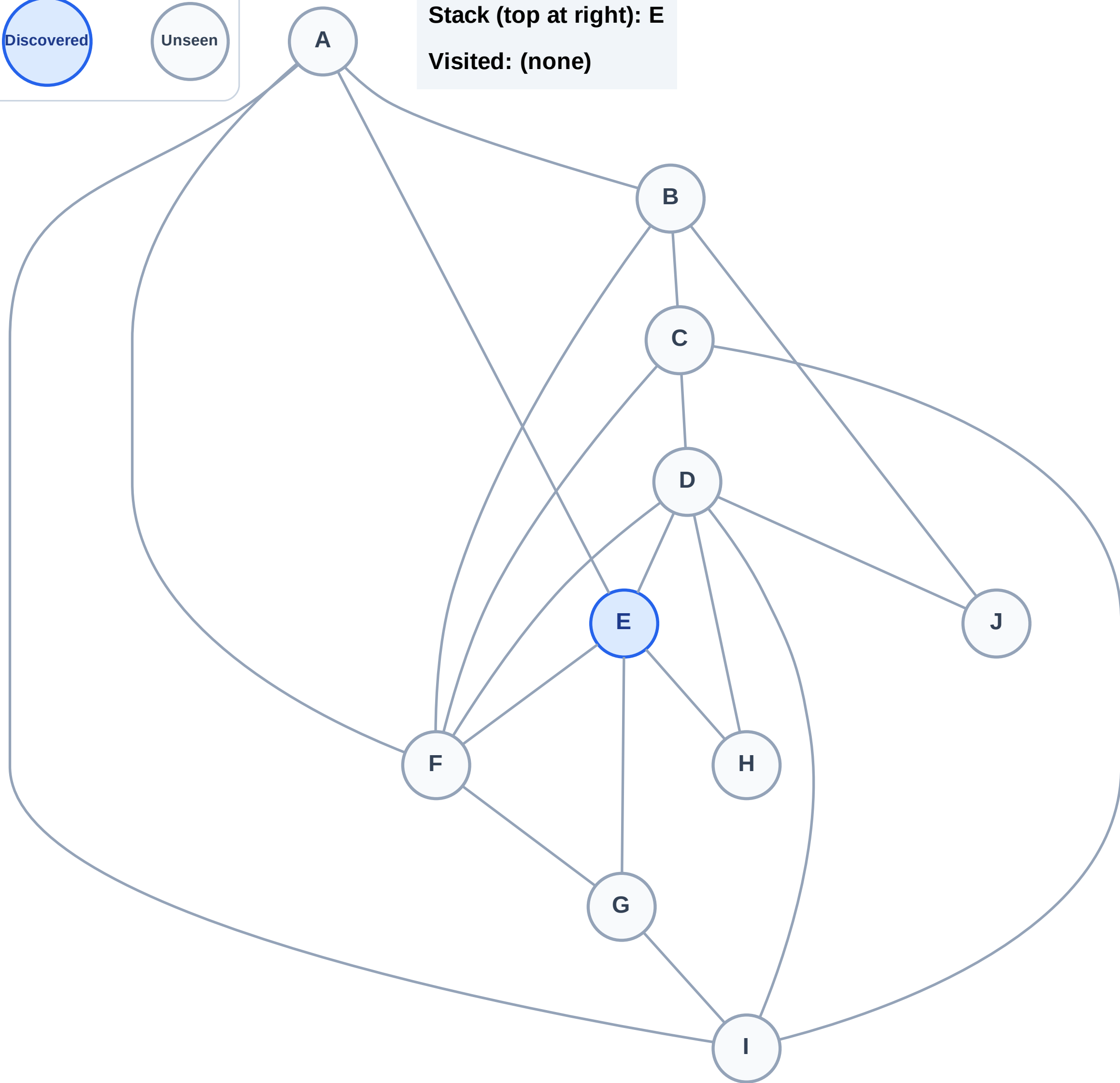
Step 1: Start at E

Legend



Stack (top at right): E

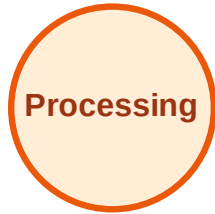
Visited: (none)



DFS Traversal

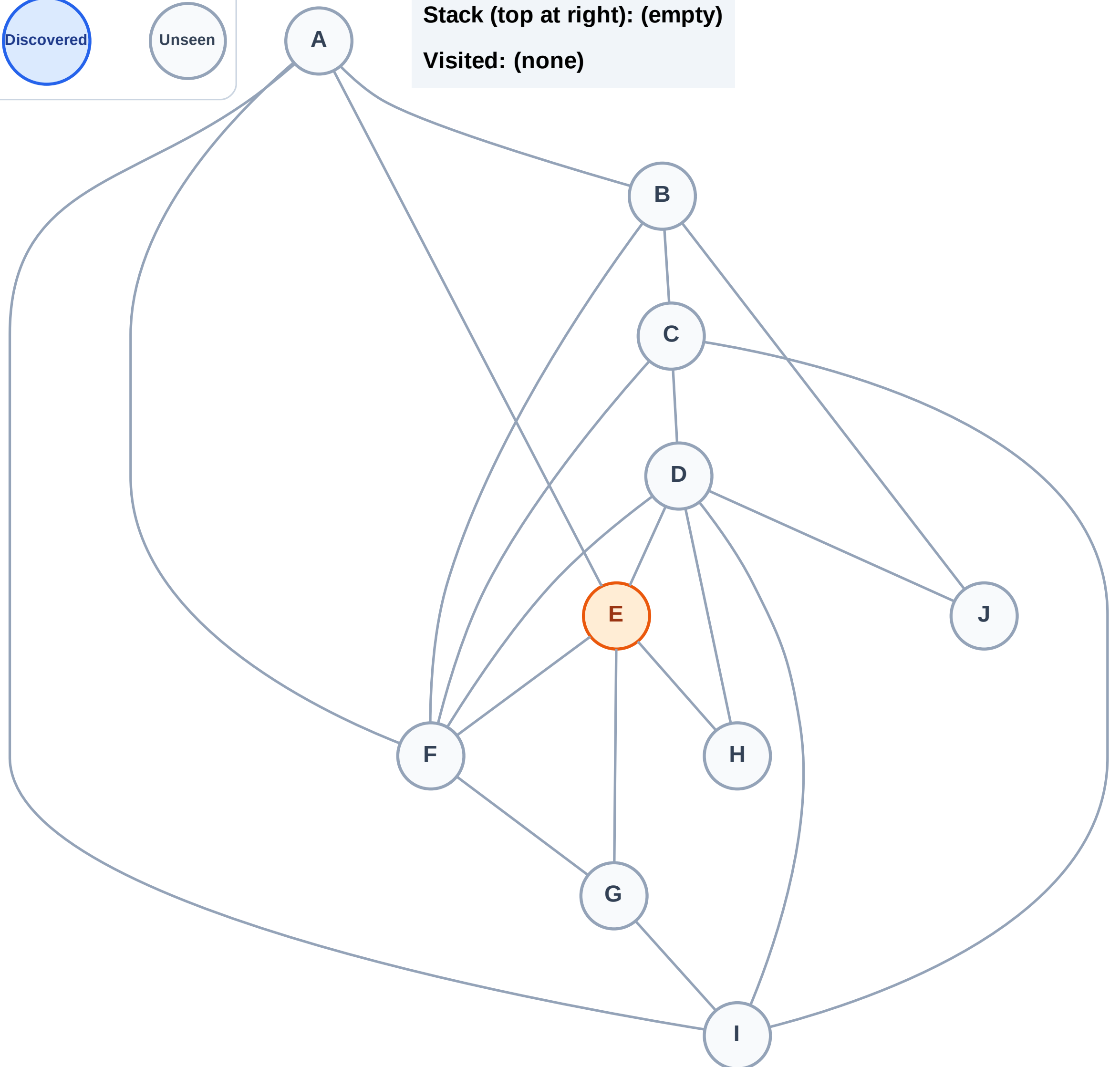
Step 2: Process node E

Legend



Stack (top at right): (empty)

Visited: (none)

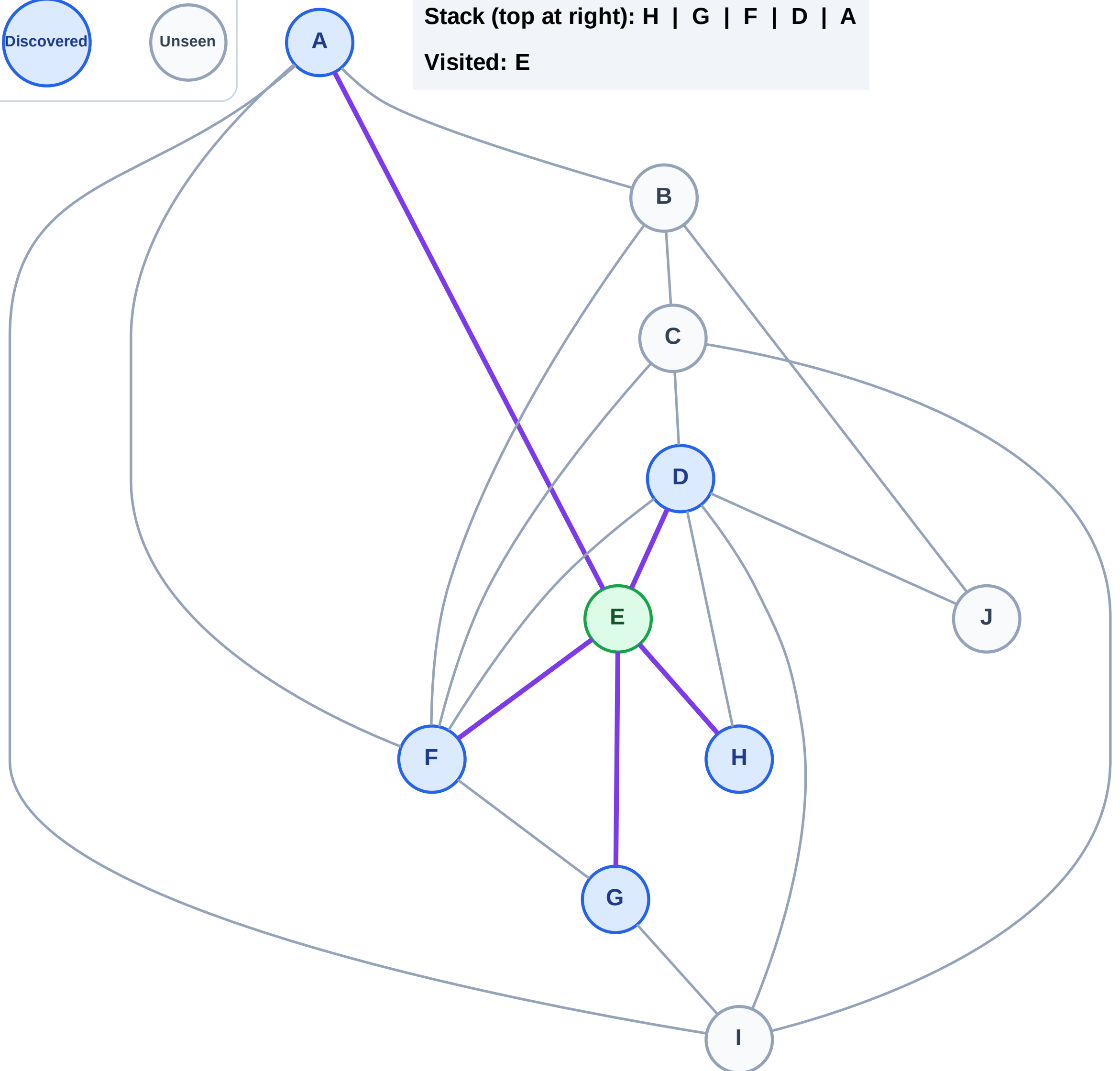


Step 3: Discover H, G, F, D, A from E

Stack (top at right): H | G | F | D | A

Legend

Unseen



DFS Traversal

Step 4: Process node A

Legend

Complete

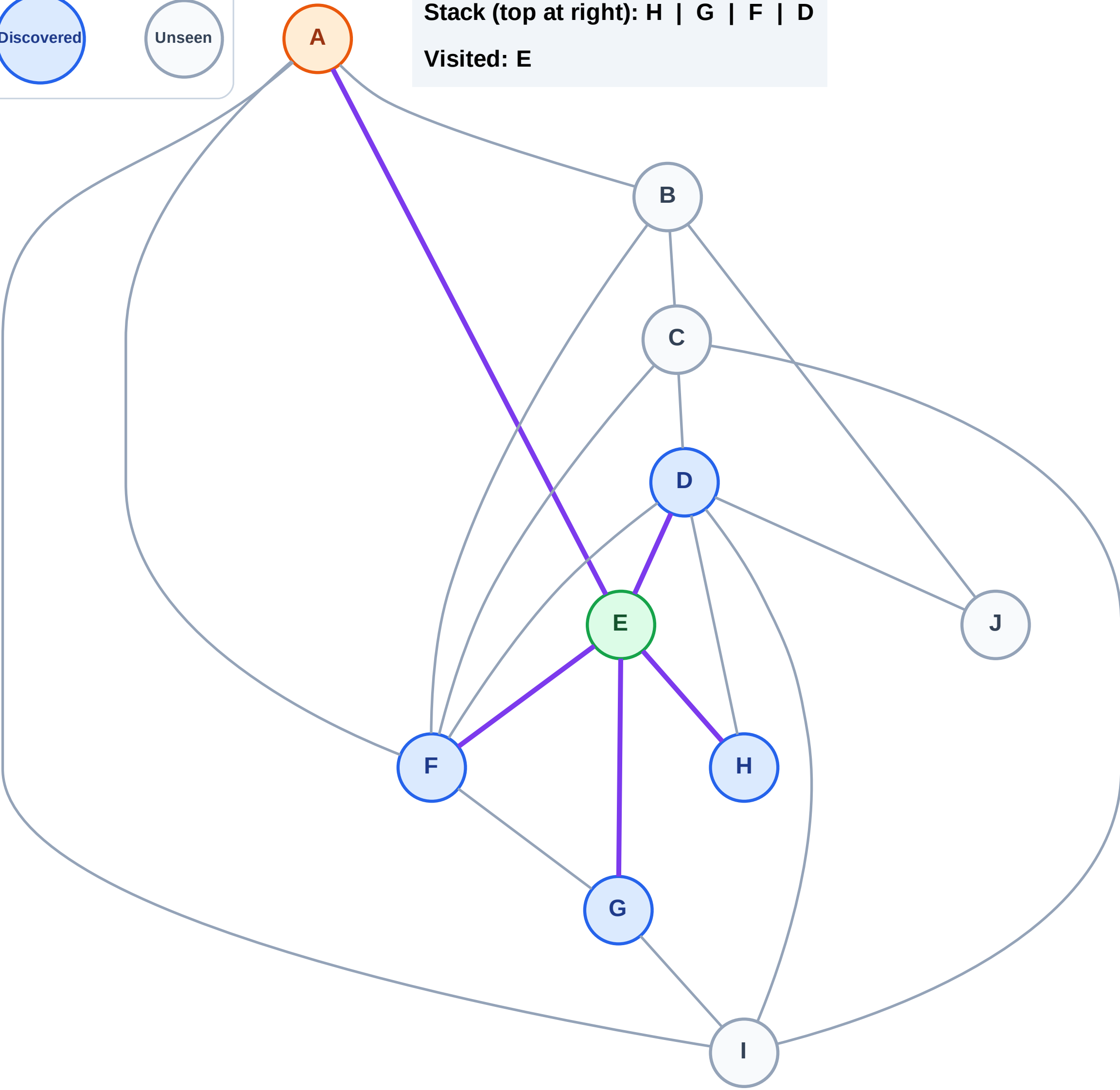
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D

Visited: E



DFS Traversal

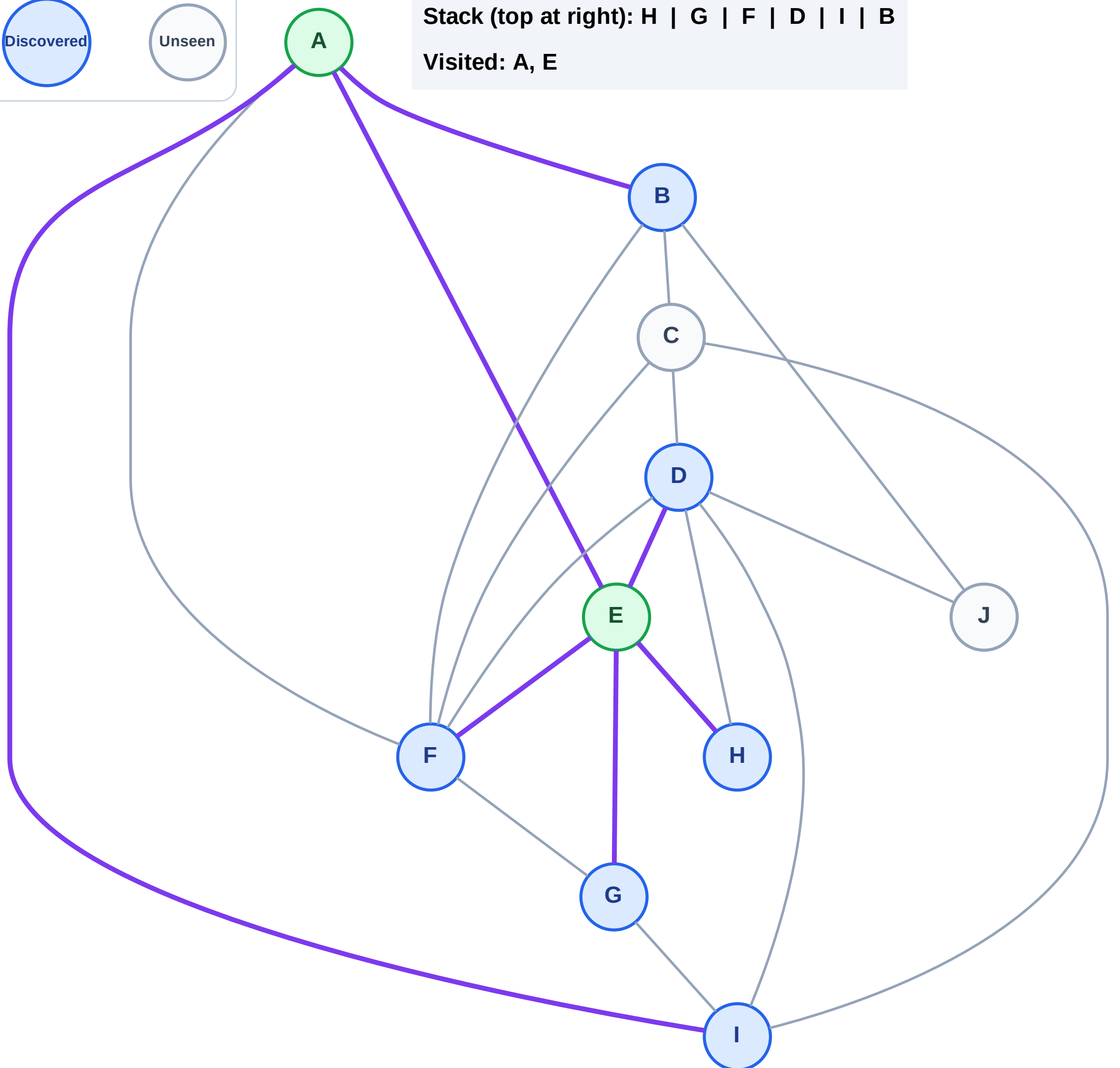
Step 5: Discover I, B from A

Legend



Stack (top at right): H | G | F | D | I | B

Visited: A, E



DFS Traversal

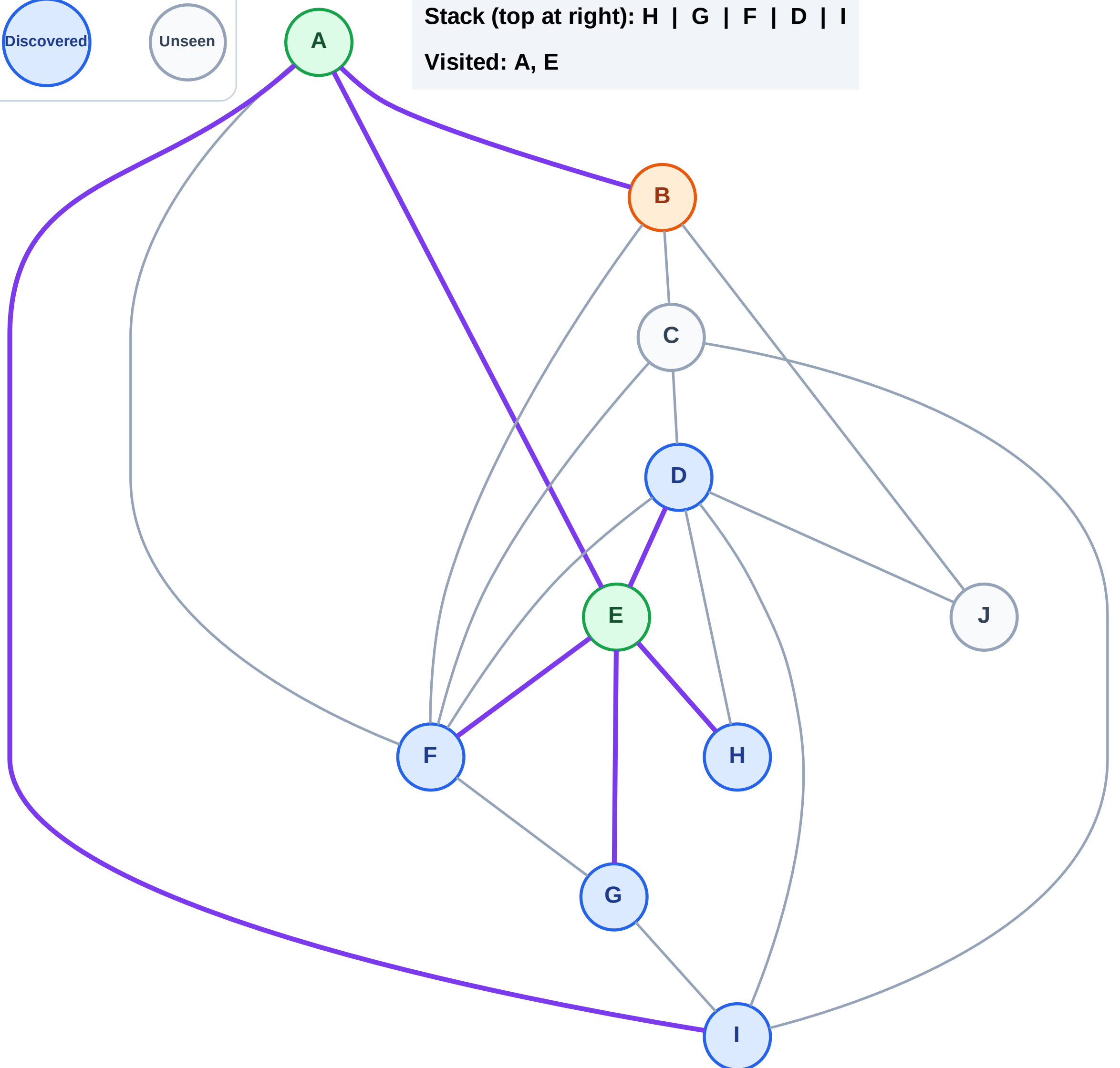
Step 6: Process node B

Legend



Stack (top at right): H | G | F | D | I

Visited: A, E



DFS Traversal

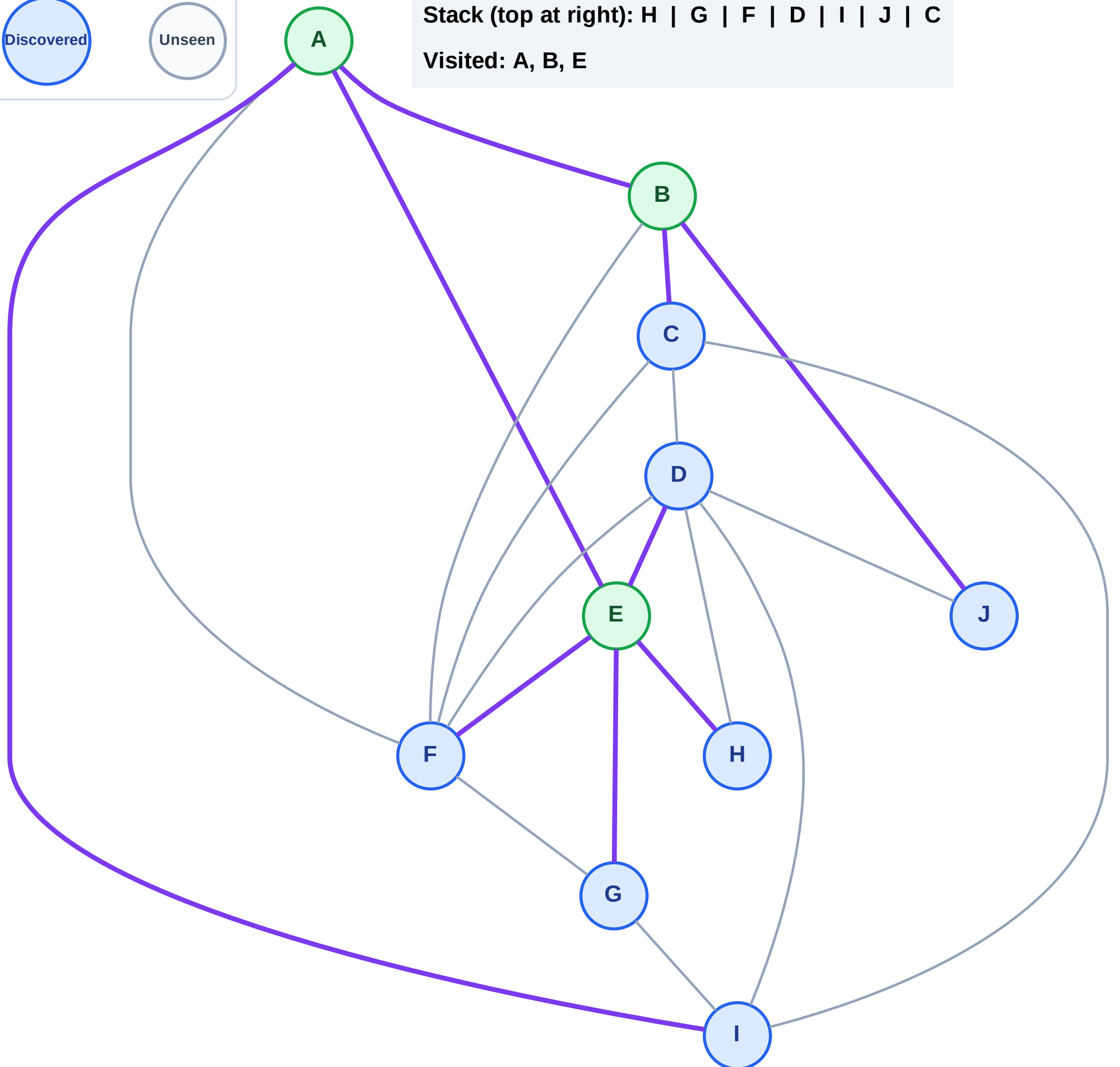
Step 7: Discover J, C from B

Legend

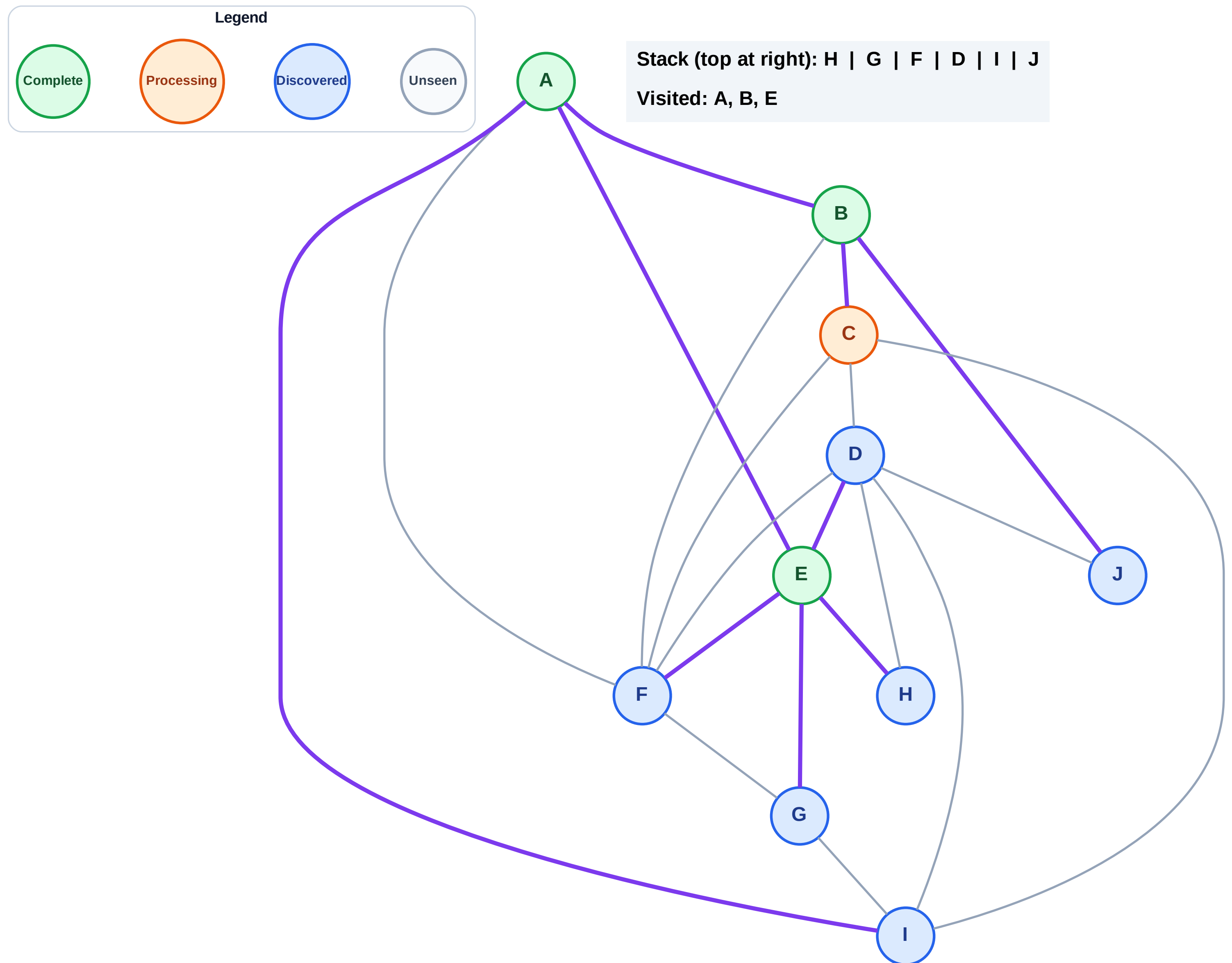


Stack (top at right): H | G | F | D | I | J | C

Visited: A, B, E



Step 8: Process node C



DFS Traversal

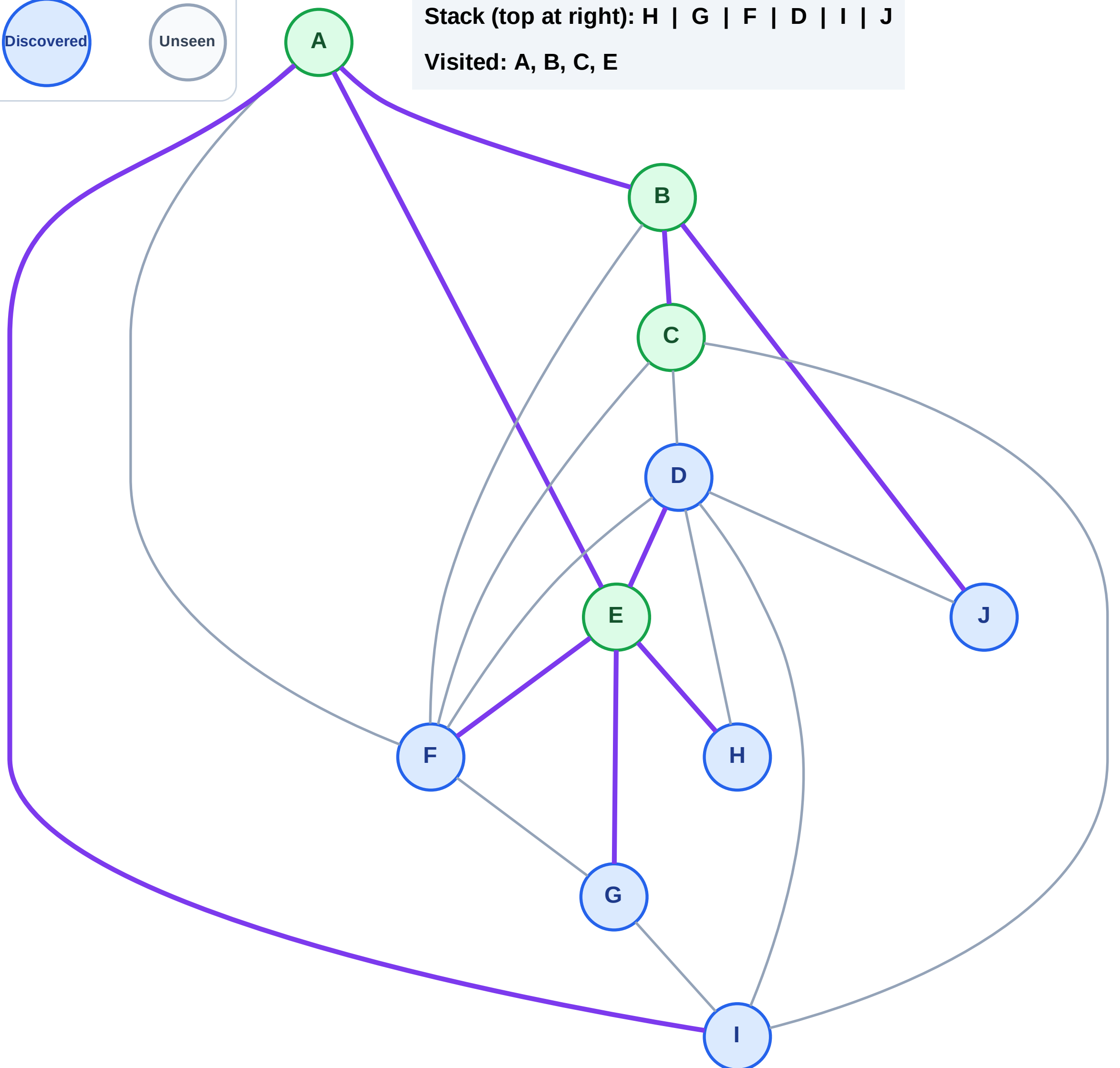
Step 9: No new neighbors from C

Legend



Stack (top at right): H | G | F | D | I | J

Visited: A, B, C, E



DFS Traversal

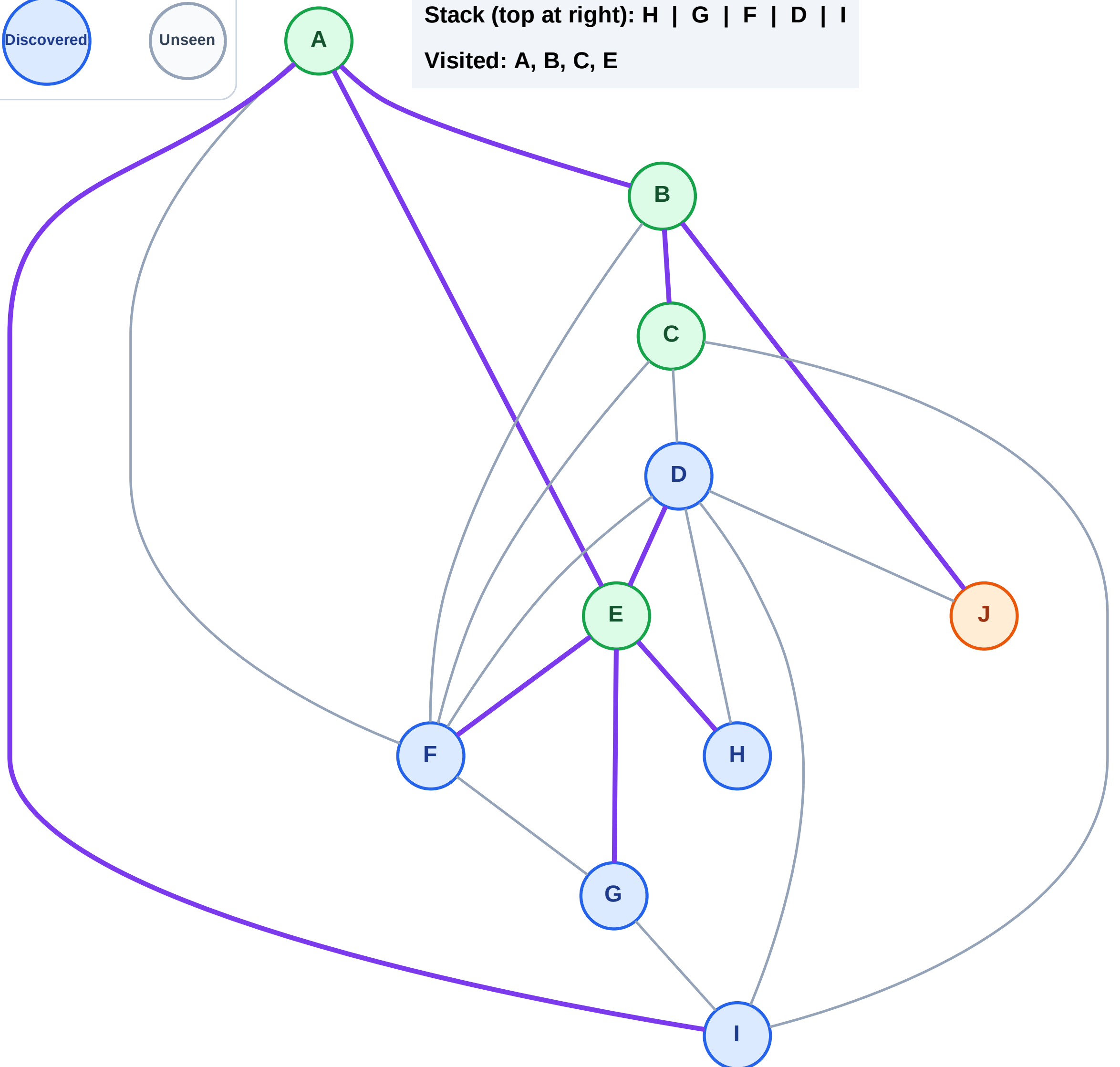
Step 10: Process node J

Legend



Stack (top at right): H | G | F | D | I

Visited: A, B, C, E



DFS Traversal

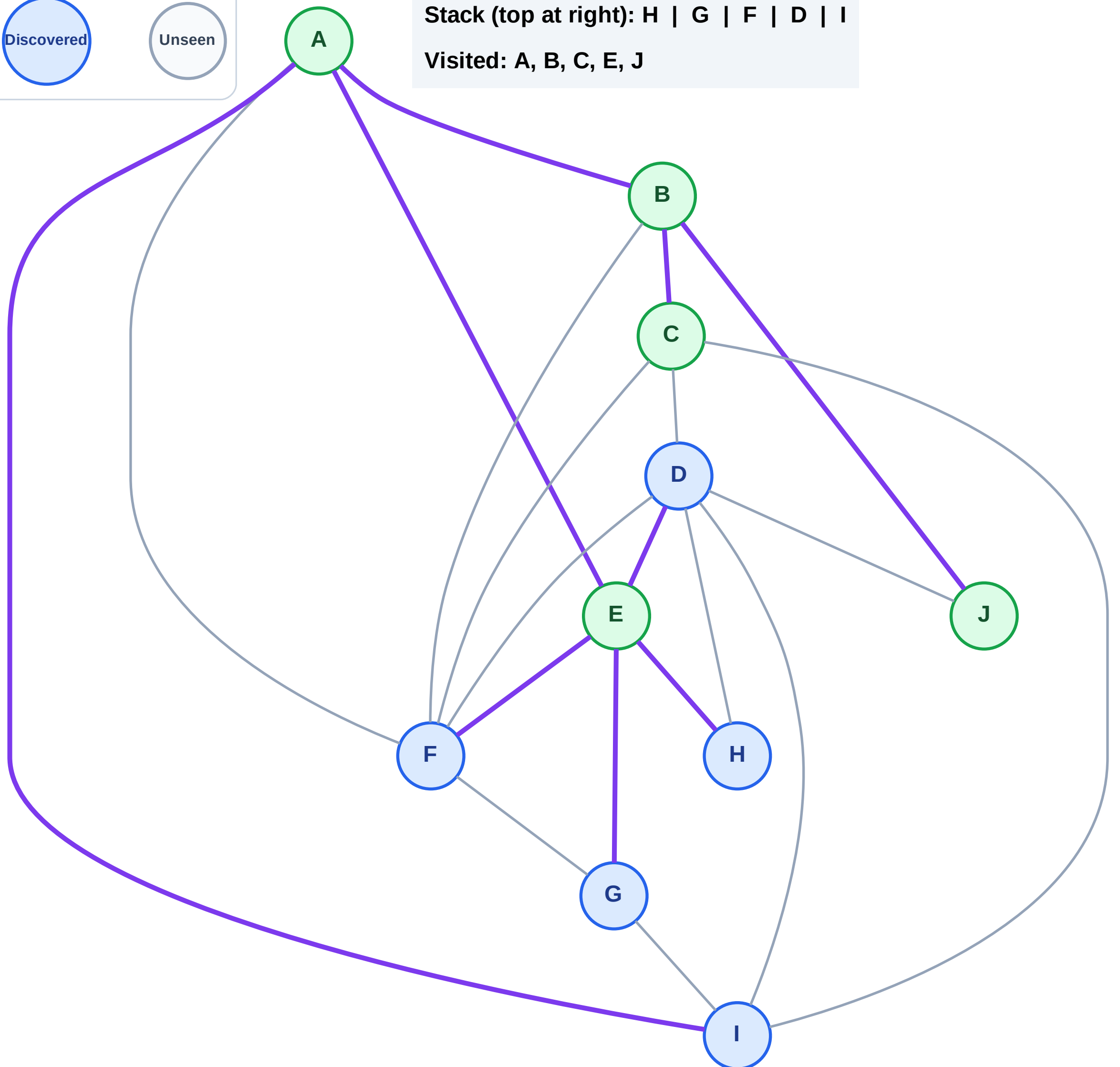
Step 11: No new neighbors from J

Legend



Stack (top at right): H | G | F | D | I

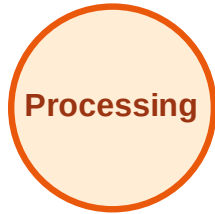
Visited: A, B, C, E, J



DFS Traversal

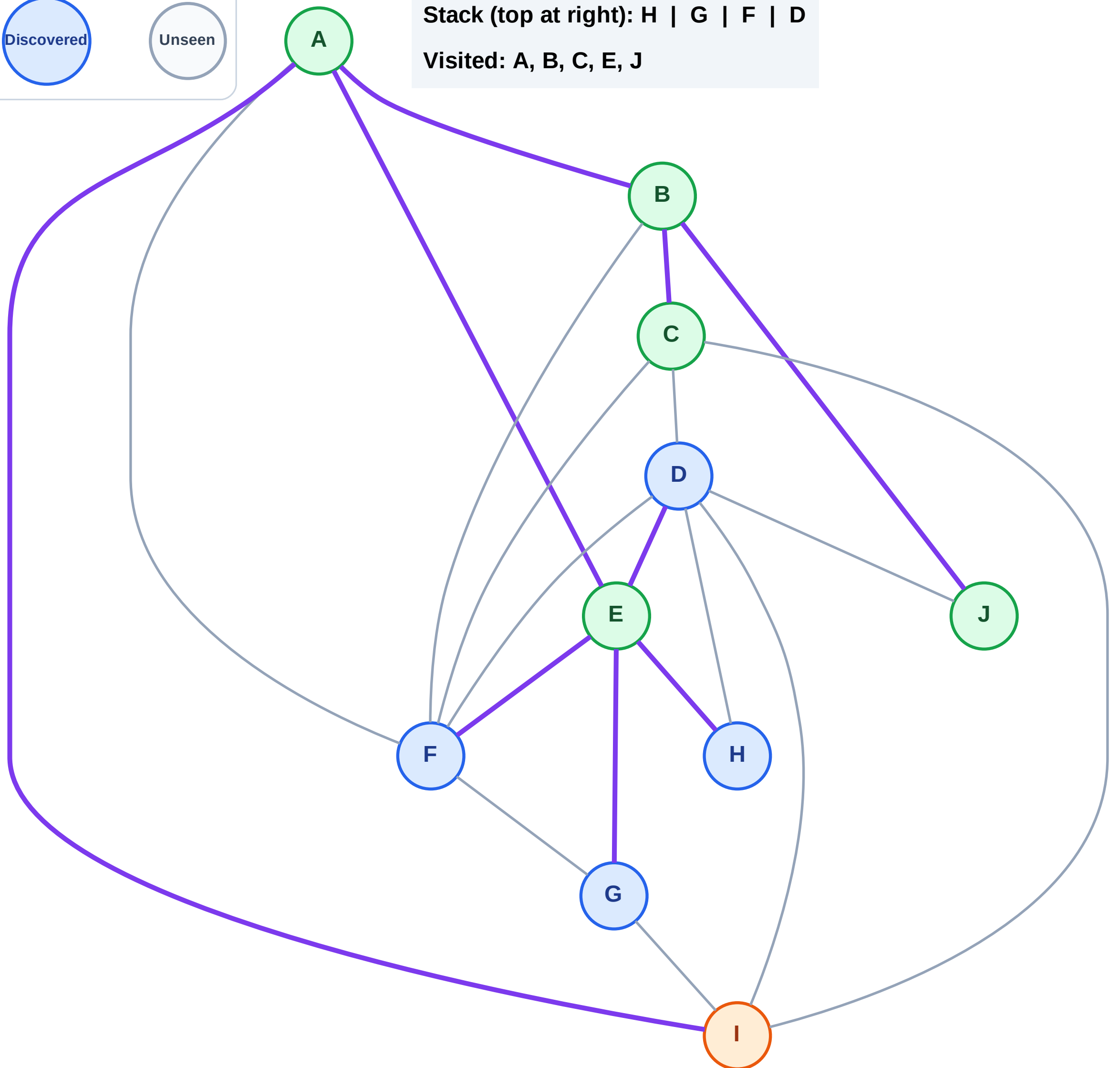
Step 12: Process node I

Legend



Stack (top at right): H | G | F | D

Visited: A, B, C, E, J



DFS Traversal

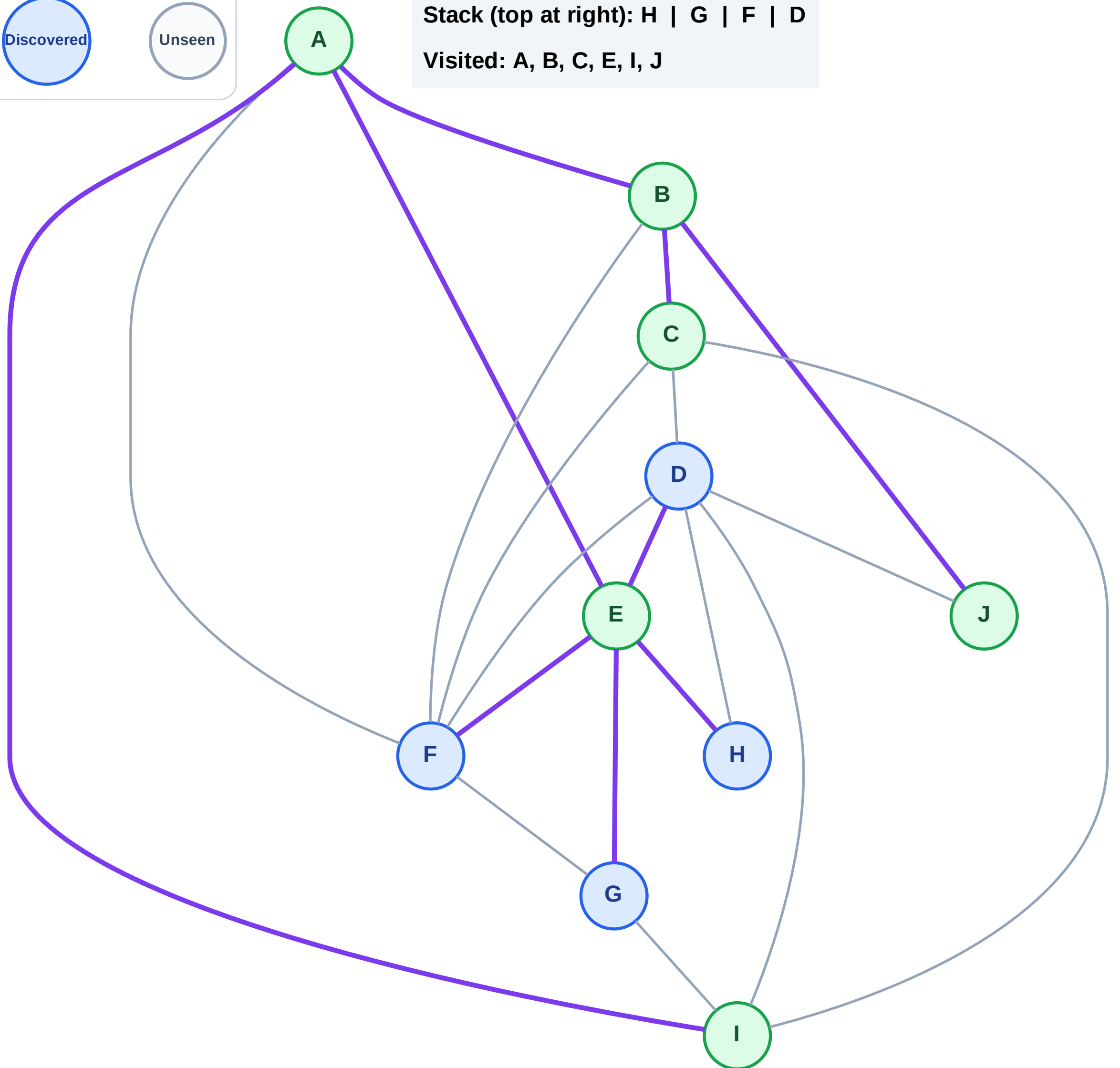
Step 13: No new neighbors from I

Legend



Stack (top at right): H | G | F | D

Visited: A, B, C, E, I, J



DFS Traversal

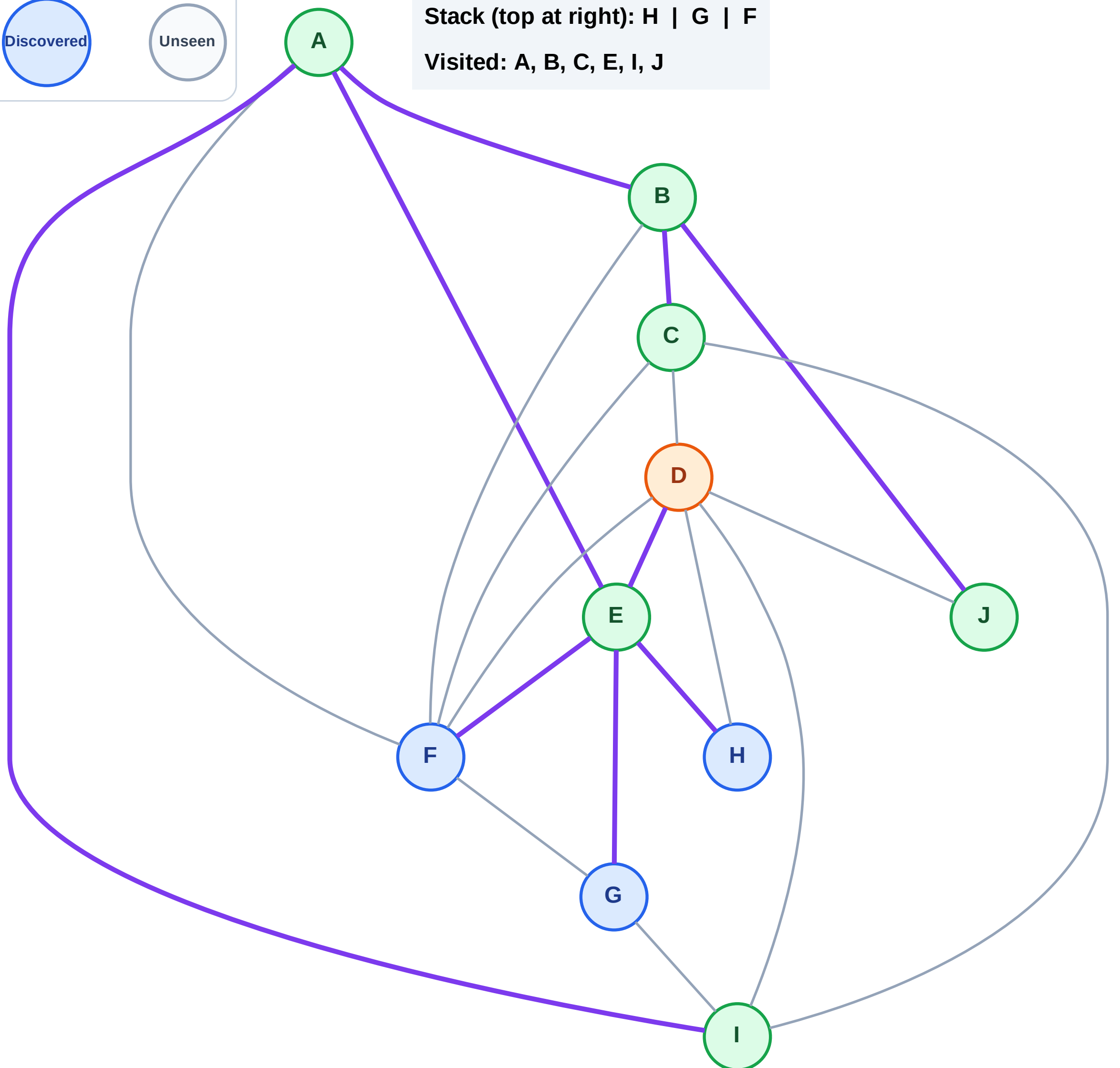
Step 14: Process node D

Legend



Stack (top at right): H | G | F

Visited: A, B, C, E, I, J



DFS Traversal

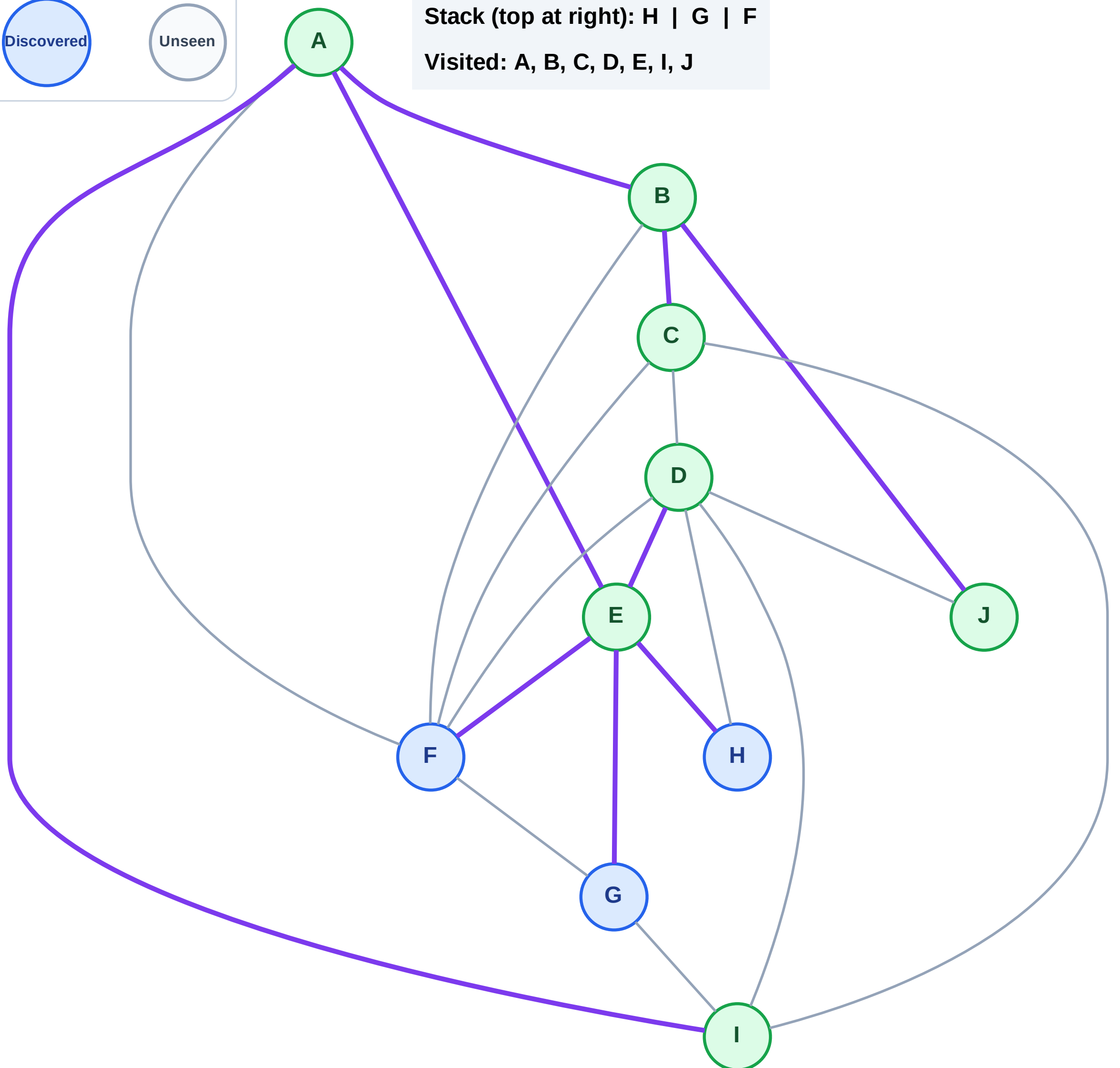
Step 15: No new neighbors from D

Legend



Stack (top at right): H | G | F

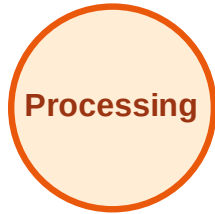
Visited: A, B, C, D, E, I, J



DFS Traversal

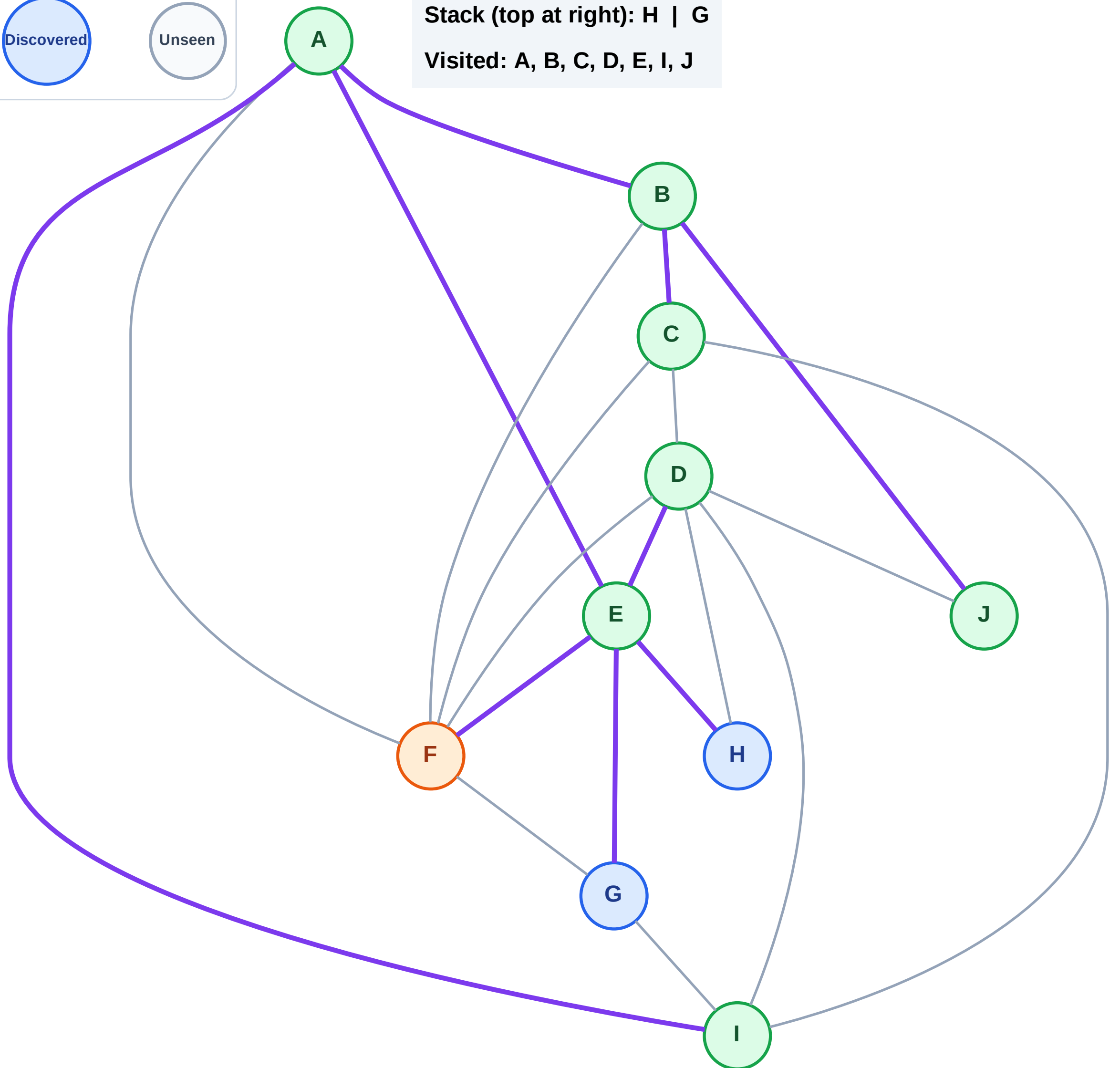
Step 16: Process node F

Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, I, J



DFS Traversal

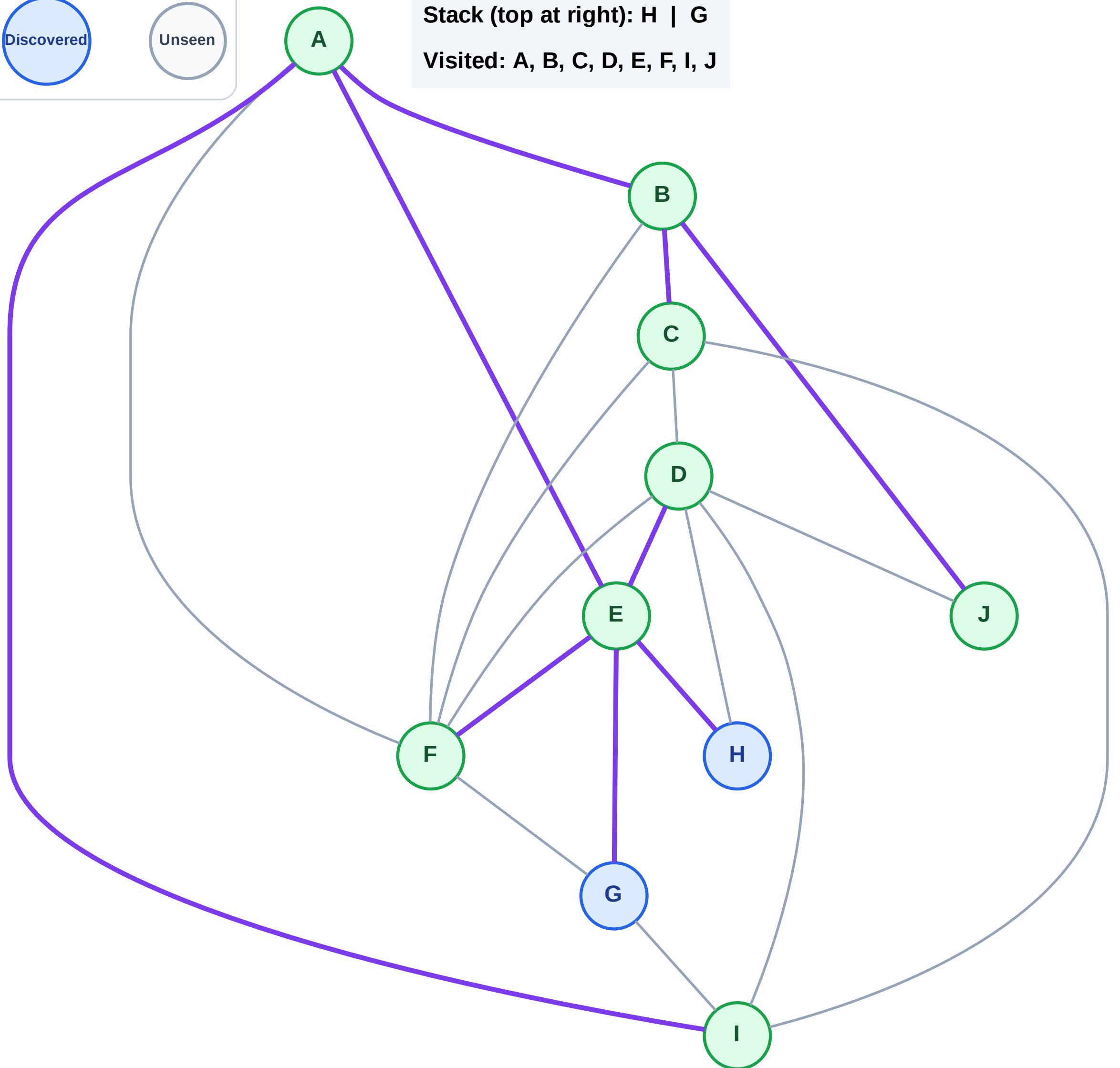
Step 17: No new neighbors from F

Legend

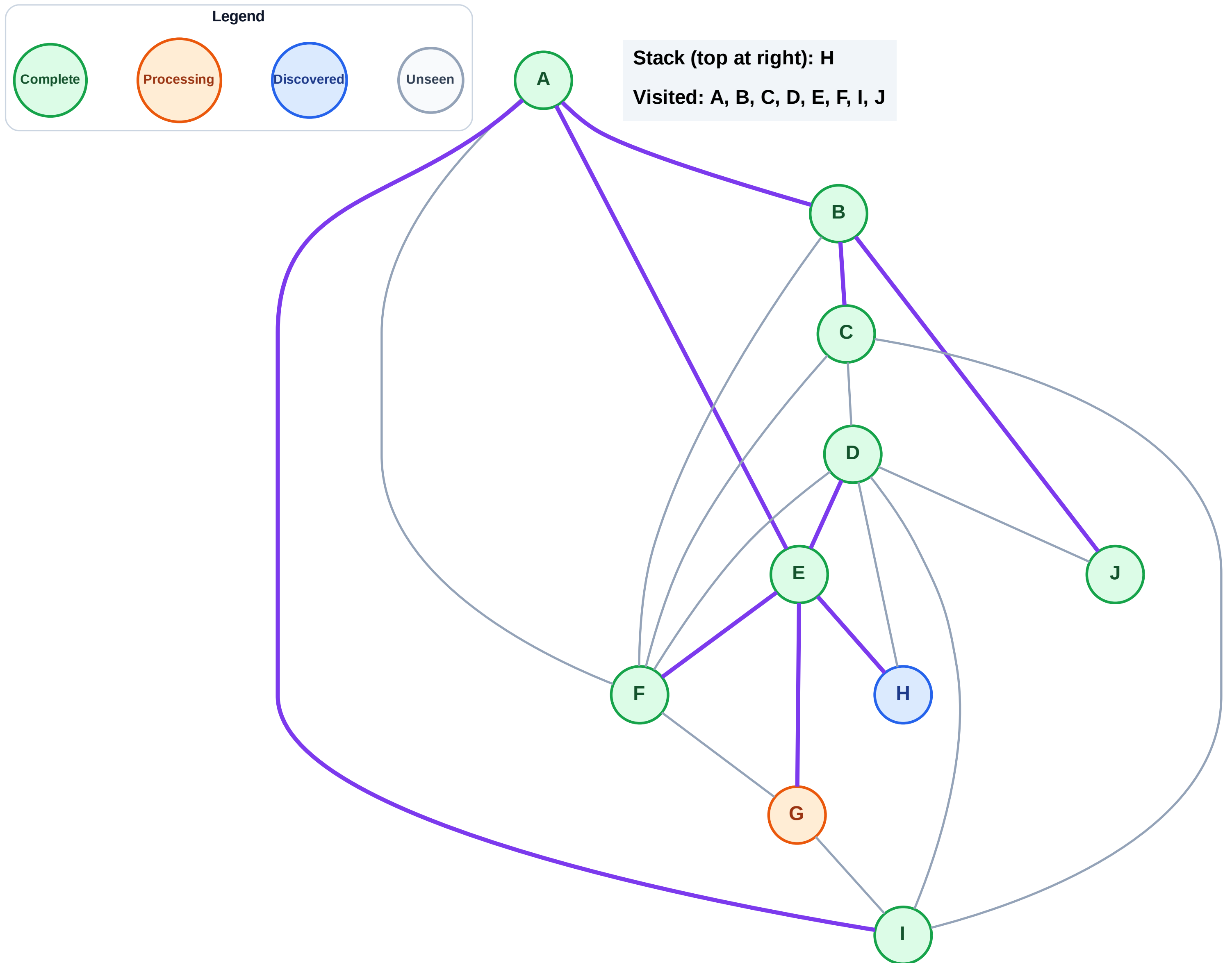


Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



Step 18: Process node G



DFS Traversal

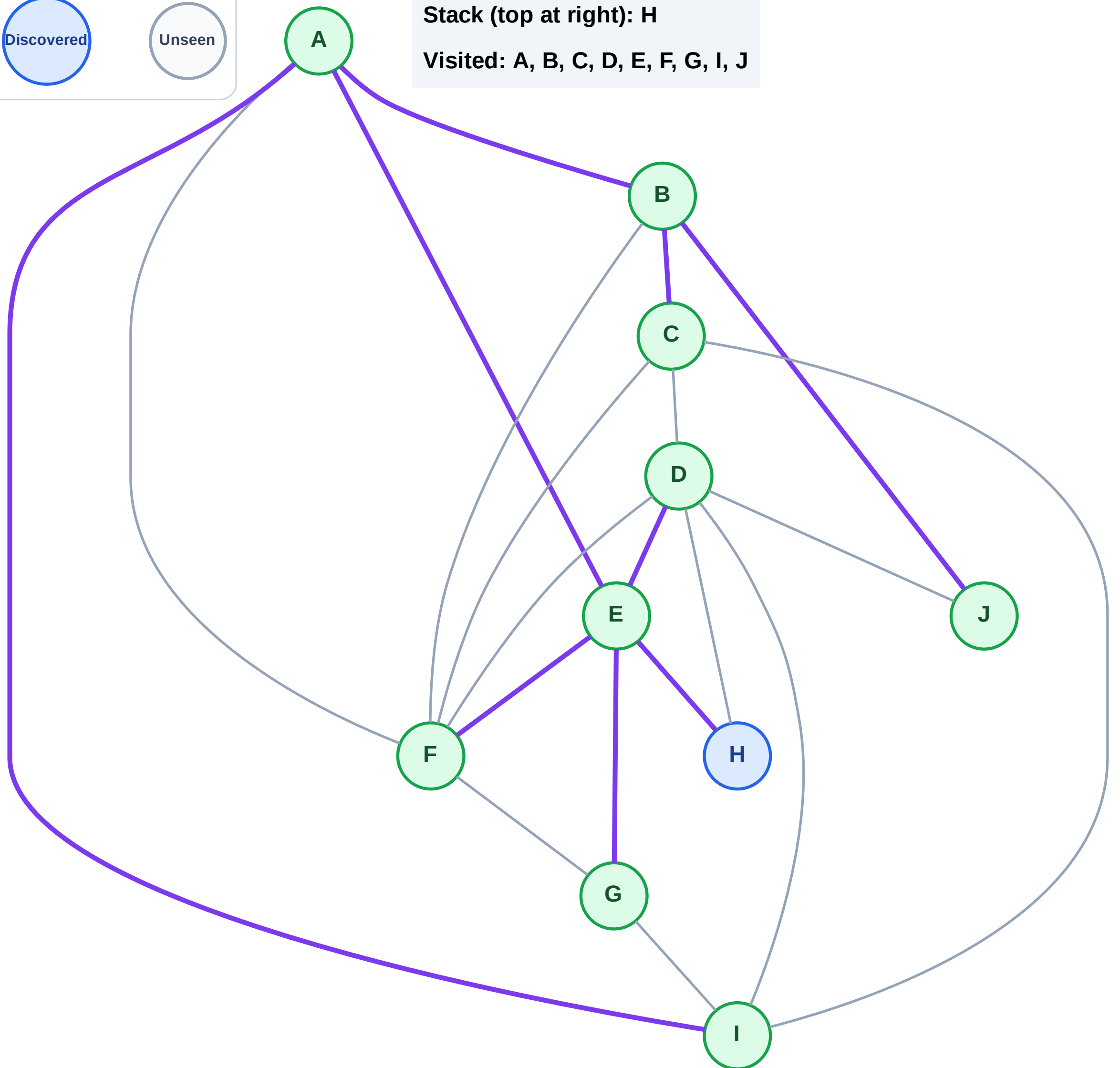
Step 19: No new neighbors from G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

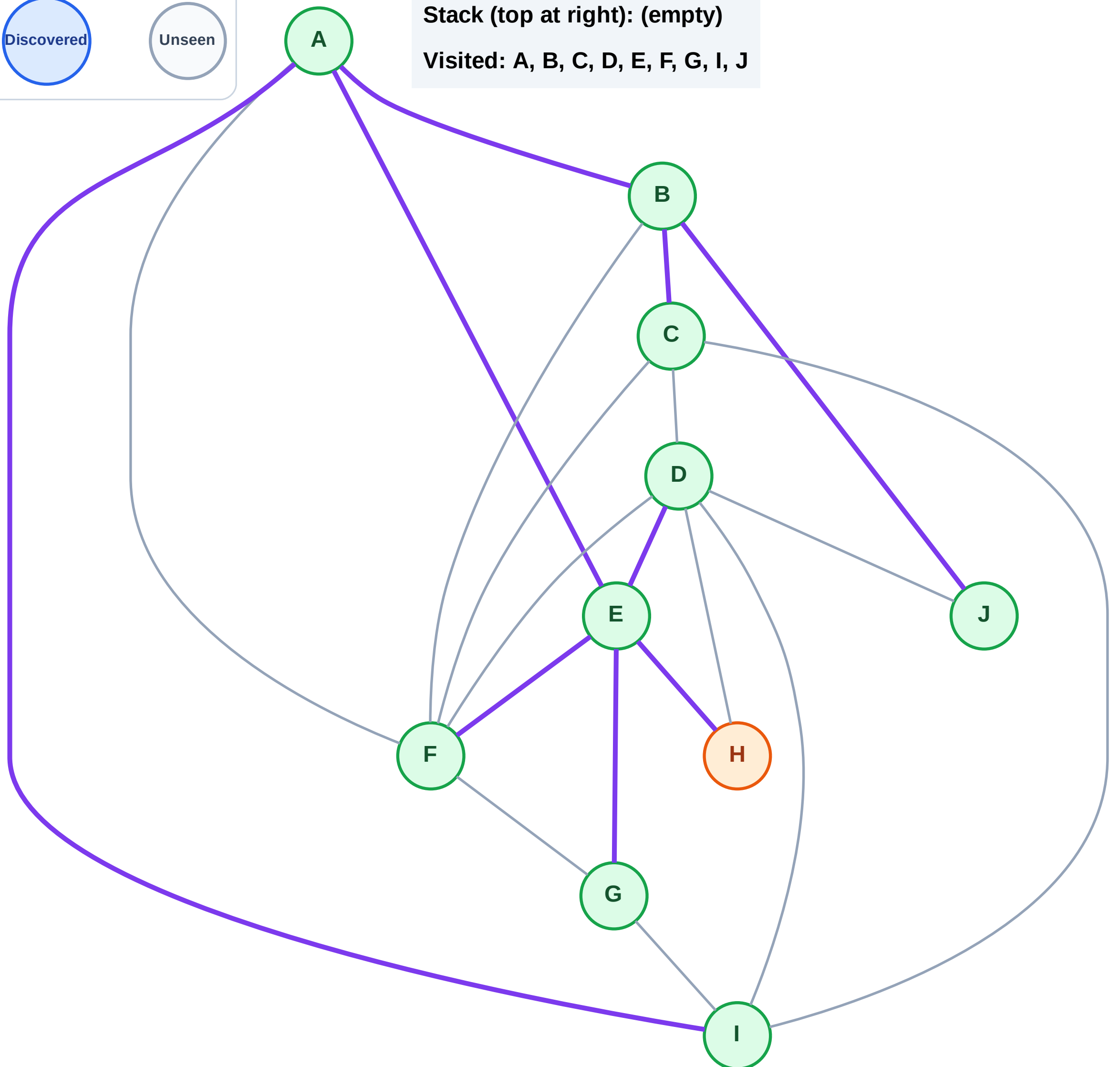
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

